Ethiopian Association of Anesthetists

Enhanced Recovery after Surgery (ERAS)

Participants' Manual

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Finally, we thank all those who supported this initiative, directly or indirectly, in advancing quality and safety of surgical and anesthesia care provided to breast cancer surgical patients through effective implementation of ERAS protocol. Your commitment to excellence has been instrumental in bringing this important resource to execution.

Introduction

Enhanced Recovery after Surgery (ERAS) programs are patient-centered, evidence-based, multidisciplinary team approaches for perioperative care to reduce the patient's surgical stress response, optimize their physiologic function, and facilitate their recovery time. This care pathway is organized in continuity starting from home, passing through different phases of prehospital/preadmission, preoperative, intraoperative, postoperative period, and back to home again. The ERAS protocols/bundles are multimodal perioperative care pathways which designed to achieve early recovery after surgical procedures by maintaining the baseline preoperative organ function (1).

Enhanced recovery after surgery (ERAS) is a paradigm shift in perioperative care, resulting in substantial improvements in clinical outcomes and cost savings. The implementation of ERAS programs involves a team commonly consisting of surgeons, anesthetists, nurses, clinical pharmacists, nutritionists, physiotherapists, ERAS coordinators and other staff from units that care for the surgical patient (2).

The ERAS Society and others promote, develop, and implement the ERAS programs, publish updated guidelines/evidence that brought dramatic changes to the perioperative care. Some of these prominent changes are promoting carbohydrate drinks about 2 hours prior to surgery than keeping the patient fast for the whole night, minimally invasive surgical approaches instead of large incisions, fluid balance rather than large volumes of intravenous fluid challenges, avoidance/early removal of catheters and drains, early mobilization, and serving of drinks and food as early as possible (3, 4).

The ERAS programs help patients to gain their functional recovery quicker while reducing hospital stays and surgical complications. The implementation of ERAS programs was found reducing perioperative complications, length of hospital stays, and readmissions; all of which have extremely significant cost implications. Despite significant amounts of evidence shows that the ERAS programs lead to improved outcomes, the implementation is too low particularly in the low- and middle-income countries due to multifactorial reasons (5).

Rationale/Justification

The implementation of the ERAS programs was evaluated by several recent studies and resulted in reduction of complication rates, length of hospital stays, costs of perioperative care, while improving patients' overall satisfaction and outcomes in varieties of surgical procedures (4, 6-12). All of these have significant implications for the health systems especially in the low-and middle-income countries such as Ethiopia. The ERAS programs are considered as the current surgical revolutions as they facilitate ready access for perioperative care by accelerating patient turnover, appropriate utilization of expensive medical resources, reduction of morbidity and mortality, and alleviating the economic burdens of healthcare. Therefore, its mandatory to implement the ERAS programs in our country and the implementation needs well-trained multidisciplinary healthcare professionals. In-spite of its benefits, ERAS programs are not being overtly implemented in our country. One of the possible barriers could be absence of well-trained healthcare professionals on the ERAS programs. The ERAS programs are new approaches with innovative paradigm shift and have not been included in pre-service training curriculum of most of healthcare professions yet. Therefore; it is essential to provide in-service training for healthcare professionals who are involved in perioperative care to ensure effective implementation of the ERAS programs. This course on ERAS programs will enable healthcare professionals to implement in their healthcare institutions effectively.

Module Syllabus

1. **Module name:** Enhanced Recovery After Surgery
2. **Module code**: ERASBC 122
3. **Module duration**: 05 days
4. **Pre-requisite**: Being healthcare professional involving in surgical care such as Surgeons, Anesthetists, Nurses, Clinical Pharmacists, Nutritionists, Physiotherapists, Operation Room Heads, Hospital and Healthcare Administrators
5. **Module continuing education unit (CEU) – 15 CEU**
6. **Module description**: This 05-days training module/course is designed for healthcare professionals who are involved in surgical care to equip with all the essential knowledge, skills and attitudes on **Enhanced Recovery After Surgery** that enable trainees to provide surgical care targeting faster functional recovery for better patient outcome.
7. **Module/course goal**: Enable healthcare professionals to effectively apply the ERAS bundles of care.
8. **Enabling/learning objectives**: After completion of this course, the trainees will be able to:
   - ✓ Discuss the ERAS bundles of care
   - ✓ Practice ERAS bundles of care
   - ✓ Develop effective leadership role in ERAS
   - ✓ Identify ERAS quality assurance during implementation
9. **Chapters/sessions outlines**:

Chapter	Title /topic	Duration (Minutes)	Sessions
1.	Basic concepts of ERAS	60	1
2.	Principles and roles of ERAS	90	2
3.	Implementation and management of ERAS	250	5
4.	The ERAS bundles of care	600	12

10. Teaching learning methods and activities:
11. Learning materials and resource (including references):
12. **Learning Assessment methods**:
13. Trainer and participants selection criteria:
14. Trainees and participants selection criteria:

15. Training evaluation methods (satisfaction, learning, behavior and impact:
16. Venue: CPD centers
17. Composition: Surgical Workforces and Clinical Oncologists

18. **ERAS Training Schedule (Day case, minimally invasive surgery)**

Time	Day 01	Day 02	Day 03	Day 04
8:30-3:30	Welcome Registration Expectations/prior experience	Preoperative Patient Education and optimization	Thromboembolism prevention	Improving Multidisciplinary communication and education
3:30-4:30	Basic concepts of ERAS (What? Why? How? When?)	Nutrition in ERAS	General and regional anesthesia in ERAS	Possible barriers to implement ERAS (Including Panel Session)
Tea Break				
4:50-5:40	Perioperative complications, functional recovery, and patient experience	Pain Management in ERAS and Opioid pandemics	ERAS in the PACU/recovery room Indwelling catheters (Why ERAS concern about them?)	Quality assurance in ERAS Closure
Lunch				
8:00-8:50	ERAS care bundles, and perioperative goals	Fluid management in ERAS	Epidural and caudal blocks	
Tea Break				

9:10-9:50	The roles of multidisciplinary healthcare professionals in ERAS	Postoperative nausea and vomiting (PONV) prevention	Chest blocks (including erector spinae block)	
9:50-10:30	Implementation and management of ERAS programs	Bowel preparation in ERAS	Abdominal wall blocks with (including quadratus lumborum block, transversalis facia block)	
10:30-10:45	Feedback time	Feedback time	Feedback time	

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Chapter One: Basic Concepts, Principles, and Roles of ERAS

Chapter duration: 170 Minutes

Chapter description: This chapter deals with basic concepts, principles, and roles of ERAS.

Chapter objective: At the end of this chapter, participants will be able to

- ✓ Discuss the basic concepts, principles, and roles of ERAS

Enabling Objective: At the end of this chapter, the participants will be able to

- ✓ Describe basic concepts of ERAS
- ✓ Explain the basic principles of ERAS
- ✓ Discuss the important roles of ERAS
- ✓ Recognize the ERAS care bundles
- ✓ Describe the perioperative goals of ERAS

Sessions

Session 1.1. Basic concepts of ERAS (What? Why? How? When?)

Session 1.2. Perioperative complications, functional recovery, and patient experience

Session 1.3. ERAS care bundles and perioperative goals

Session 1.1. Basic concepts of ERAS

(What? Why? How? When?)

Session duration: 90 min

Session objectives

- ✓ Describe the basic concepts of ERAS
- ✓ Discuss importance roles of ERAS in perioperative care

1.1.1. Introduction

Activity: Small group discussion

Discuss basic concepts of ERAS

Duration: 5 min

Enhanced Recovery after Surgery (ERAS) Society consensus statements and guidelines are powerful tools that have been implemented across hospitals and healthcare systems worldwide to improve the quality of surgical care. The approach was initially conceived by a group of surgeons in northern Europe, based on the principle that actions undertaken to modulate postoperative stress can improve recovery after surgery. The ERAS approach has shown that early mobilization, early reintroduction of nutrition and rapid discharge are feasible and beneficial for a large number of postoperative patients.

The first ERAS consensus paper was produced for colonic surgery in 2005. a team of expert clinical researchers and translated an understanding of physiological responses to surgical stress into clinical practice, with a focus on the specific needs of adults undergoing colorectal surgery. The guideline was produced by experts, based on a joint review of the literature and the quality of evidence supporting individual aspects of the guideline.

Its expansion to other settings and countries resulted in multiple studies showing benefits in patients undergoing colorectal surgery. Subsequent guidelines were developed using modifications of the colorectal ERAS approach and applied to various other surgical specialties, with excellent results

Enhanced recovery after surgery (ERAS) models are; a multi-modal perioperative care pathways. They are designed to achieve early recovery after surgical procedures by maintaining preoperative organ function and reducing the profound stress response following surgery.

Activity: Small group discussion and gallery walk

Discuss and present basic concept, elements and importance of ERAS

Duration: 15 minutes

Instruction to conduct gallery Walk

- ✓ Provide the assignment one day before the session
- ✓ Prepare the space
- ✓ Prepare the materials
- ✓ Introduce the activity
- ✓ Divide into groups
- ✓ Allow time for discussion
- ✓ Wrap up

1.1.2. Elements of ERAS in perioperative care

The elements of ERAS typically include:

- ✓ Pre-operative optimization: Patients are assessed for any underlying health issues and treated accordingly before surgery. They are also educated about the surgery and recovery process.

- ✓ Peri-operative care: Patients are given appropriate pain management and anesthesia to minimize post-operative pain and discomfort. They are also encouraged to move and breathe deeply as soon as possible after surgery to prevent blood clots and lung infections.

- ✓ Nutrition: Patients are given a high-protein diet before and after surgery to promote healing and reduce muscle wasting. They are also encouraged to drink fluids and eat as soon as possible after surgery.

- ✓ Avoiding unnecessary interventions: Unnecessary procedures, such as nasogastric tubes and urinary catheters, are avoided to reduce the risk of complications.

- ✓ Early mobilization: Patients are encouraged to get out of bed and move around as soon as possible after surgery to prevent blood clots and lung infections.

- ✓ Discharge planning: Patients are discharged as soon as they are stable and able to care for themselves at home, with clear instructions and appropriate follow-up care to ensure a smooth recovery.

1.1.3. Basic corner stone of ERAS

There are six cornerstones in ERAS From the initial consultation with the surgeon to the month-long follow-up visit. The main cornerstones of ERAS are;

- Evidence based perioperative care
- Multi professional and multi-disciplinary approach
- Team work
- Continuous interactive audit and reporting
- Data driven change
- Readiness to make the next change

Evidence based perioperative care ERAS care is based on the information that is available in the medical literature that aim to find information that can help improve the outcome for patients undergoing surgery. Experts in the field are expected to review and grade the knowledge in the medical literature in systemic way and build an evidence-based guide

Multi professional and multi-disciplinary approach the ERAS team is the core of having the ERAS in place in a hospital unit. Because it is a completely new and different way to run care, it has to have the full support of the management/administration, heads of departments and other decision makers. All professionals and specialties need to be represented on the team to ensure successful implementation of the ERAS protocols. The team should secure that there is at least one member covering every unit engaged in the care of the patient. this include surgeons anesthesiologist, pain management focal person, nurse, physiotherapists, and dietician.

The team collects key data from patents and meet regularly and make decisions to align their local practices to make sure adherence to the guidelines

Teamwork is an essential component of ERAS, as it involves the coordination and collaboration of multiple healthcare professionals, including surgeons, nurses, anesthesiologists, and physical therapists. The goal of teamwork in ERAS is to ensure that patients receive consistent, high-quality care throughout the entire surgical process, from pre-operative preparation to post-operative recovery. This includes clear communication, shared decision-making, and the use of evidence-based best practices. By working together effectively, the healthcare team can help to optimize patient outcomes and improve the overall surgical experience for patients.

Continuous interactive audit and reporting in ERAS refers to the ongoing monitoring and evaluation of a patient's progress throughout the entire recovery process using interactive tools and data analysis. This can include tracking vital signs, monitoring for complications, and evaluating the effectiveness of various interventions. The goal of continuous interactive audit and reporting is to ensure that patients are receiving the best possible care, and that any issues are identified and addressed as soon as possible.

Data-driven change in ERAS involves using data and analytics to continually assess and improve the program, such as by monitoring patient outcomes and identifying areas for improvement. This can include analyzing data on patient demographics, comorbidities, surgical procedures, and outcomes to identify trends and potential areas for improvement in the program. Additionally, data-driven change in ERAS can involve using machine learning algorithms to predict patient outcomes, identify high-risk patients, and optimize the care provided to patients.

To make the next change in an ERAS program, it is important to conduct a thorough review of the current protocol and identify areas that can be improved. This may involve gathering data on patient outcomes, surveying patients and staff, and consulting with experts in the field. Once potential changes have been identified, it is important to test them in a pilot study before implementing them on a larger scale. It is also important to monitor the results of the changes and make adjustments as needed to ensure that the ERAS program continues to improve patient outcomes.

Activity: group discussion Describe cornerstones of ERAS in perioperative care 10 minutes

1.1.4. Benefits of the Enhanced Recovery Programs

Compared to traditional care, Enhanced Recovery programs have been shown to be safer and more effective.

Decreased length of stay (LOS) /Faster recovery times: Patients who participate in ERAS typically experience shorter hospital stays and are able to return to normal activities more quickly than those who do not. Many studies have shown that, in addition to a shorter length of stay, the use of Enhanced Recovery protocols is associated with lower or constant morbidity and mortality.

Reduced postoperative pain: The ERAS program incorporates various strategies to minimize pain and discomfort following surgery, such as the use of nerve blocks and minimally invasive techniques

Another important aspect of Enhanced Recovery protocols is their ability to either accelerate or prevent the decline of organ function.

Fewer complications: The ERAS program emphasizes the importance of preoperative education, risk assessment, and evidence-based practices, which can help to reduce the risk of complications such as infection, blood clots, and pneumonia.

Reduced readmissions The ERAS protocol also shines when it comes to reducing the rate of readmission following surgery.

Reduced costs The ERAS program can also lead to cost savings for hospitals and payers by reducing the length of hospital stays, decreasing the risk of complications, and improving patient outcomes.

Improved patient experience improved patient satisfaction: Patients who participate in ERAS often report higher levels of satisfaction with their surgical experience, as they feel more in control of their recovery and have a better understanding of what to expect

Activity: individual reflections How could Enhanced Recovery Programs reduce the coast? 15 minutes

Summary

ERAS Society consensus statements and guidelines are powerful tools that have been implemented across hospitals and healthcare systems around the world to improve the quality of surgical care. Early mobilization, early reintroduction of nutrition and rapid discharge have been shown to be feasible and beneficial for a large number of postoperative patients. ERAS has basic principles for reducing recovery that are as follows: optimization prior to surgery, Postoperative care Nutrition, keeping unnecessary interventions to a minimum: avoid Procedures that are unnecessary, Early mobilization and discharge planning: for effective implementation and outcome

The program should follow the ERAS pillars of "evidence-based perioperative care," Multi-professional and multidisciplinary approaches, teamwork, interactive auditing, and reporting on a continuous basis Data-driven transformation and availability to make the next change

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Session 1.2: Perioperative complications, functional recovery and patient experience

Session duration: 70 Minutes

Session objective: At the end of this chapter, participants will be able to

- ✓ Describe common perioperative complications
- ✓ Discuss the patterns of functional recovery of surgical patients
- ✓ Explain patient experience during the perioperative period

1.2.1. Introduction

Activity: Reflection on prior experience

How was patient recovery in your clinical setup?

Time: 10 minutes

Perioperative complications refer to any adverse events that occur during the process of preparing for, undergoing, and recovering from surgery. These complications can range from minor issues such as infection or bleeding, to more serious complications such as organ failure or death. Some common perioperative complications include: hemorrhage, infection, and embolism, anesthetic complications, cardiac or respiratory complications, wound healing problems, adverse reactions to medications, nerve/tissue damage, and organ failure. It is important for healthcare professionals to closely monitor patients before, during, and after surgery in order to detect and treat any complications that may arise.

The occurrence of perioperative complications significantly affects the functional recovery of surgical patients and regaining the baseline physiologic status. Unsuitable recovery has substantial impacts on morbidity, mortality, length of hospital stays, cost of healthcare, and service delivery. All of these have significant implications on patients' and caregivers' experience during the perioperative period.

The implementation of ERAS programs has reduced the occurrence of perioperative complications, improved functional recovery, and patients' experience during the perioperative period.

1.2.2. Common perioperative complications

Surgical patients can develop multi-systemic complications during the perioperative period. These complications range from common but minor self-limiting complaints to rare but serious disabling morbidities and death. The occurrence and consequences of perioperative complications depend on different factors such as patient factors, human factors, and setting factors. Clinicians are continuously introducing innovative strategies and protocols to reduce the occurrence of these complication. The implementation of ERAS is the current paradigm shift in the perioperative care which is improving patient recovery. Recent evidences have shown that ERAS has reduced the incidence of perioperative complications, hospital stay, and readmission due to postoperative complications compared to the conventional care (1-3).

Activity: Small group discussion and presentation

How can ERAS reduce the occurrence of this complications?

Time: 10 minutes

Common perioperative complications:

- ✓ Surgical site (wound) infection;
- ✓ Anastomotic leak;
- ✓ Pain
- ✓ Intestinal ileus;
- ✓ Nausea and vomiting;
- ✓ Deep venous thrombosis;
- ✓ Respiratory complications (pneumonia, hypoxia, hypoventilation…);
- ✓ Cardiac complication (hypo/hypertension, arrhythmias, myocardial ischemia/infarction…);
- ✓ Postoperative cognitive dysfunctions;
- ✓ Surgical mortality

The implementation of ERAS bundles of care during the perioperative can reduce these complications and enhance surgical outcomes.

1.2.3. Functional recovery of surgical patient

Hospitalization and surgery in often lead to a loss of strength, mobility, and functional capacity. Patients need to regain their baseline functional capacity as early as possible after surgery. Functional recovery refers to the process of regaining the ability to perform normal daily activities after an injury, illness or medical condition. It involves physical, cognitive and emotional rehabilitation to improve mobility, strength, coordination, and overall functioning. The goal of functional recovery is to improve the individual's quality of life and independence. The return to normal activities can take anywhere from a few days to several weeks or months.

Postoperative Quality of Life (QoL) can be described in terms of a person's physical, emotional, and social well-being after surgery. It can be affected by various factors such as pain, mobility, independence, and return to daily activities. It is important to assess postoperative QoL to evaluate the success of a surgical procedure and guide future treatment decisions. Traditionally, surgical outcomes measured by minor and major complications, hospital length of stay and sometimes longer-term survival. Each of these is important but there needs to be greater emphasis on patient-reported outcome measures. Global measures of a patient's quality of recovery, avoidance of postoperative morbidities, early hospital discharge to home (without re-admission) and longer-term disability-free survival can better define postoperative recovery.

A patient's recovery pathway can be mapped through the immediate days or weeks after surgery with documentation of morbidity using the postoperative morbidity survey and/or a quality of recovery score, days alive and at home up to 30 days after surgery and then longer-term disability-free survival using the WHO Disability Assessment Schedule 2.0 scale (**Annex 1**). These can be used to define quality of recovery after surgery.

1.2.4. Patient experience

Activity: Brief group discussion

How was patients' experience about their perioperative care at your healthcare facility?

What were the most common complaints?

Time: 5 minutes (3 minutes for discussion and 2 minutes for reflection)

The perioperative patient experience refers to the events that occur before, during and after a surgical procedure, including the patient's physical, emotional and psychological well-being. It encompasses the patient's interaction with healthcare providers, pain management, and the quality of care received, outcomes of care, and the overall satisfaction with the surgical experience. Improving the perioperative patient experience can lead to better outcomes, promoting patient well-being, functional recovery, increased patient satisfaction, and improved patient safety.

Patient experience can be affected by multiple factors such as communication, received care/treatment, and outcomes. To ensure better patient experience, clinicians should provide patient centered care, proactively take action to ensure patient satisfaction, measure and monitor patient experience and feedback, and take action on patients' feedback. Providing excellent patient experience requires effort and dedication. Valuing patient experience is fundamental for a successful perioperative care.

Enhanced recovery after surgery (ERAS) is a patient-centered approach to care that aims to improve patient outcomes and experience by reducing stress, optimizing function, and minimizing postoperative complications. Implementation of the ERAS bundles of care such as preoperative education, clear communication, optimized nutrition, pain management, PONV prevention early mobilization, and minimizing the use of catheters and drains. The results of these efforts are improved patient outcomes, reduced length of hospital stays, improved patient satisfaction, and better overall patient experience.

Activity: Critical appraisal on current evidences and reflections

How was patient experience comparing the traditional/conventional approaches and ERAS?

Time: 15 minutes

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Session 1.3. ERAS care bundles and perioperative goals

Session duration: 60 Minutes

Session objective: At the end of this session, participants will be able to;

- ✓ Describe ERAS bundles of care
- ✓ Recognize perioperative goals of ERAS

1.3.1. Introduction

Activity: Small group discussion and reflections

What are the ERAS bundles of care?

Time: 15 minutes

Enhanced Recovery After Surgery (ERAS) is a protocol for improving patient outcomes and reducing the length of hospital stays following surgical procedures. The ERAS bundle is a set of evidence-based guidelines aimed at optimizing patient care before, during, and after surgery. The exact components of an ERAS bundle may vary depending on the surgical specialty and patient population. The key is to implement a tailored, evidence-based program that meets the specific needs of each patient and surgical procedure.

1.3.2 ERAS care bundles

The ERAS bundles of care;

1. Preoperative patient education
2. Optimize nutrition
3. Optimized management of comorbid medical conditions
4. Preoperative carbohydrate loading and prevent dehydration
5. Reduced fasting times before surgery
6. Optimal fluid management
7. Effective pain management (multimodal approach)
8. Minimized use of perioperative opioids

9. Early initiation of oral feeding
10. Early mobilization, physical therapy, and rehabilitation
11. Reduced use of catheters, drains, and other devices that can slow recovery
12. Safe and effective postoperative care

Implementing an ERAS bundle requires a multidisciplinary approach, involving collaboration between surgeons, anesthesiologists, nurses, and other healthcare providers. Regular assessment and data collection are also important components of an ERAS program, to ensure continuous improvement and ensure that the program remains effective over time.

1.3.3 Perioperative goals of ERAS

The perioperative goals of ERAS include:

1. Enhance quality of recovery after surgery
2. Reduce length of hospital stay
3. Reduce perioperative complication
4. Minimize healthcare cost
5. Improve patient outcome
6. Improve patient satisfaction

Another important aspect of ERAS is patient involvement and empowerment. Patients are encouraged to take an active role in their own care and recovery, through preoperative preparation and education, and ongoing communication and support from their healthcare team.

The ERAS programs are patient-centered approach to surgical care that aims to improve outcomes and enhance recovery by following a set of evidence-based guidelines. The overall aim of ERAS is to reduce postoperative complications and speed up recovery, allowing patients to return to their normal activities as soon as possible.

Implementing an ERAS program requires a thorough evaluation of existing practices and the development of a customized plan to meet the specific needs of each surgical specialty and patient population. The success

of an ERAS program is dependent on strict adherence to the guidelines and regular reassessment to ensure continued improvement.

Activity: Small group discussion and reflections

What are the ERAS bundles of care?

What are the perioperative goals of ERAS?

Time: 20 minutes

Summary

- The ERAS bundle is a set of evidence-based guidelines aimed at optimizing patient care before, during, and after surgery.
- The Implementing an ERAS bundle requires a multidisciplinary approach, involving collaboration between surgeons, anesthesiologists, nurses, and other healthcare providers.
- The ERAS programs are patient-centered approach to surgical care that aims to improve outcomes and enhance recovery.
- The success of an ERAS program is dependent on strict adherence to the guidelines and regular reassessment to ensure continued improvement.

Chapter Two: Implementation, challenges and quality assurance in ERAS

Chapter duration: 150 Minutes

Chapter description: This chapter deals with the implementation and management of ERAS programs

Chapter objective: At the end of this chapter, participants will be able to

- ✓ Identify the appropriate modalities and barriers of implementing and managing ERAS programs

Enabling Objective: At the end of this chapter, the participants will be able to

- ✓ Describe the roles of multidisciplinary healthcare professionals in ERAS
- ✓ Discuss implementation approaches of ERAS programs
- ✓ Elaborate management principles of ERAS programs
- ✓ Develop effective multidisciplinary communication and education in ERAS teams
- ✓ Discuss possible barriers to the implementation of the ERAS
- ✓ Recognize quality assurance modalities in ERAS

Sessions

Session 2.1. The roles of multidisciplinary healthcare professionals in ERAS

Session 2.2. Implementation and management of ERAS programs

Session 2.3. Improving Multidisciplinary communication and education

Session 2.4. Possible barriers to the implementation of the ERAS programs

Session 2.5. Quality assurance in ERAS

Session 2.1. The roles of multidisciplinary healthcare professionals in ERAS

Session duration: 40 Minutes

Session Objectives: At the end of this session participants will be able to:

- ✓ List roles of surgeons in ERAS
- ✓ Identify roles of anesthetists in ERAS protocol
- ✓ List roles of Nurses in ERAS
- ✓ Identify roles of pharmacists in ERAS
- ✓ List roles of dieticians in ERAS

2.1.1. Introduction

Activity: Small group discussion and reflection

What is the importance of mentioning the roles of health professionals in ERAS

Time: 10 minutes

Enhanced recovery after surgery (ERAS) protocols are multimodal perioperative care pathways designed to achieve early recovery after surgical procedures by maintaining pre-operative organ function and reducing the profound stress response following surgery. Different health professionals do have different roles or responsibilities for the implementation of ERAS. Everybody's role is complimentary each other for the whole ERAS protocol.

2.1.1.1. Roles of Surgeon:

The roles of a surgeon in an Enhanced Recovery after Surgery (ERAS) program include:

- ✓ Preoperative planning: Assessing patient eligibility and discussing the ERAS protocol with them.
- ✓ Surgical technique: Adhering to the principles of minimally invasive surgery and reducing tissue trauma.

- ✓ Perioperative care: Providing evidence-based care during the perioperative period to minimize stress and promote recovery.
- ✓ Postoperative management: Monitoring the patient's progress and managing complications as needed.
- ✓ Patient education: Educating the patient about the ERAS protocol and postoperative care.
- ✓ Collaboration: Working with a multidisciplinary team of healthcare professionals to ensure the best outcomes for the patient.

2.1.1.2. Roles of the Anesthetists:

An Anesthetists in an ERAS program is responsible for providing anesthesia care for patients undergoing surgery. This includes preoperative assessment and evaluation, intraoperative monitoring and management of anesthetic techniques, and postoperative pain management. An Anesthetist may also provide emergency care for patients in the ER, such as intubation and resuscitation. In addition, anesthetists may be involved in the development and implementation of ERAS protocols and protocols for other surgical specialties.

2.1.1.3. Roles of Nurse:

The roles of a nurse in an ERAS (Enhanced Recovery after Surgery) program include:

- ✓ Preoperative patient assessment and education
- ✓ Management of pain and discomfort
- ✓ Administering medications and treatments
- ✓ Monitoring vital signs and fluid balance
- ✓ Assisting with mobility and physical therapy
- ✓ Maintaining accurate patient records and communication with the healthcare team
- ✓ Assisting with discharge planning and postoperative care instructions.

2.1.1.4 Roles of Pharmacist:

Pharmacists play several important roles in the Enhanced Recovery after Surgery (ERAS) protocol, including:

- ✓ Medication management: Pharmacists ensure that patients receive the correct medications at the right time, with proper dosing and administration, to support the ERAS protocol.
- ✓ Pain management: Pharmacists work with the healthcare team to develop and implement a pain management plan that minimizes the use of opioids and other potentially harmful medications.
- ✓ Nutrition support: Pharmacists collaborate with dietitians to ensure that patients receive appropriate nutritional support to promote healing and recovery.
- ✓ Fluid management: Pharmacists help to monitor fluid balance and electrolyte levels to ensure that patients receive the appropriate amount of fluids to support recovery.
- ✓ Education: Pharmacists provide education to patients, families, and other healthcare providers about the ERAS protocol and the medications used to support it.
- ✓ Monitoring and evaluation: Pharmacists monitor patients' progress and evaluate the effectiveness of the ERAS protocol and the medications used to support it, making adjustments as needed.

2.1.1.5. Roles of Dietitians:

Dietitians in ERAS programs play an important role in helping patients to adopt healthier eating habits before, during, and after surgery. They provide nutrition counselling and education to help patients understand the importance of following a healthy diet and lifestyle before and after surgery. Dietitians also help create individualized meal plans that meet the patient's nutritional needs and provide guidance on how to shop for and prepare healthy meals. Additionally, they may provide guidance on how to manage any dietary restrictions or intolerances that the patient may have in general.

Activity: Role play

1. Make a group of 6 and 5 of you be assigned as surgeon, Anesthetist, Nurse, pharmacist and the 6th one be a patient.
2. Mention and debrief your roles to each other.
3. Introduce yourself to the patient and do your roles.

Time: 20 Minutes

Summary

ERAS, Enhanced Recovery after Surgery, is an approach to improving surgical outcomes by optimizing the patient care pathway. It involves engaging a team of health professionals to ensure that pre-operative preparation, intraoperative care, and post-operative recovery are managed safely and effectively. Attached painting is a representation of this coordinated effort, with multiple professions working in tandem to provide the best possible patient experience. The painting is called "The Collaboration". Coordination is required between members of the perioperative period to ensure successful implementation of ERAS protocols, including the surgical team, anesthesia team, nursing staff, preoperative, intraoperative, and postoperative personnel.

Session 2.2. Implementation and management of ERAS programs

Session duration: 90 Minutes

Session objectives: At the end of this session participants will be able to:

- ✓ Discuss the steps of implementation process of ERAS

2.2.1. Introduction

Activity: Small group discussion and gallery walk

Describe implementation steps of ERAS

Time: 30 min

Instruction to conduct gallery Walk

- ✓ Provide the assignment 10 minutes before the session
- ✓ Prepare the space
- ✓ Prepare the materials
- ✓ Introduce the activity
- ✓ Divide into groups
- ✓ Allow time for discussion
- ✓ Wrap up

Successful ERAS implementation depends on many factors, including patient and staff education, engagement, collaborative leadership culture between the facility and interdisciplinary team, and use of outcome data to continuously improve elements of the pathway. Successful change management and ERAS implementation occurs as a process that evolves from creation of an engaging story, leadership, a climate supportive of the change initiative, implementation or trial of the pathway in one surgical specialty, and sustained change with continued improvement. Despite the proven successes of ERAS, implementation challenges exist

due to long held team member and patient opinion of how things should be done. In order to clearly implement the program, the health facility or team should follow the steps below

Conduct a thorough assessment of current practices and identify areas for improvement: Review current literature and guidelines on Enhanced Recovery after Surgery (ERAS) to ensure adherence to best practices. Gather data on patient outcomes, including length of stay, complications, and readmission rates. Assess the role of nutrition in recovery and assess the adequacy of preoperative fasting and postoperative feeding.

Develop a multidisciplinary team to plan and implement the program: a multidisciplinary team should include surgeons, anesthesiologist, physical therapist, dietitian and social worker. It can be beneficial to have representation from the following areas:. -Pharmacists - to ensure appropriate medication management. Social worker - to address social and financial needs. Case manager - to coordinate care between hospital and home.

Clear protocols and guidelines for patient care in enhanced recovery after surgery (ERAS): typically include the following components: Preoperative preparation, Intraoperative care, Post-Surgery Care, Pain Management and Discharge Planning. Continual audit and improvement of the ERAS is required with monitoring of the outcomes

Implement evidence-based practices to optimize patient outcomes Incorporate: patients and their families in the decision-making process to improve patient outcomes. Use guidelines and clinical practice recommendations from reputable sources such as the World Health Organization. Continuously assess and improve the overall quality and safety of care provided by your healthcare provider.

Monitor progress and continuously evaluate the program's effectiveness: to monitor progress and evaluate the effectiveness of an enhanced recovery after surgery program, the following steps can be taken:

- ✓ Collect data on patient outcomes, such as length of stay, complications, and readmissions
- ✓ Compare the data to pre-program benchmarks and national standards
- ✓ Assess patient satisfaction through surveys
- ✓ Continuously review and modify the program based on the data and feedback
- ✓ Regularly review the program with the multidisciplinary team to identify areas for improvement and make adjustments as needed.

Provide ongoing education and training for all team members: there are several ways to provide ongoing education and training for team members in an enhanced recovery after surgery (ERAS) program:

- ✓ Regular in-service training sessions: These can be held regularly, such as monthly or quarterly, to provide updates on new evidence, guidelines, and best practices for ERAS.
- ✓ Online learning resources: These can include webinars, online courses, and educational modules that team members can access at their convenience.
- ✓ Journal clubs: Team members can review and discuss relevant articles on ERAS to stay up-to-date on the latest research.
- ✓ Mentorship and shadowing: Team members can observe and learn from experienced practitioners who are already implementing ERAS in their practice.
- ✓ Conferences and workshops: Team members can attend relevant conferences and workshops to learn from experts in the field and network with other practitioners.
- ✓ Team-building and communication strategies: Regular team meetings and communication strategies can help to ensure everyone is working towards the same goal and can share ideas and best practices.

Encourage patient and family involvement in the program

- ✓ Educate patients and families on the importance of their participation in the program.
- ✓ Involve them in goal setting and decision making.
- ✓ Provide clear communication and updates on the patient's progress.
- ✓ Encourage self-care and provide resources for at-home recovery.
- ✓ Involve them in planning for discharge and follow-up care.
- ✓ Acknowledge and address any concerns or questions they may have.
- ✓ Empower them to take an active role in their recovery

Use data and feedback to make ongoing adjustments and improvements to the program. To use data and feedback to make ongoing adjustments and improvements to an enhanced recovery after surgery program, the following steps can be taken:

- ✓ Collect data on patient outcomes, such as length of stay, complications, and readmissions.
- ✓ Use this data to identify areas where the program may be falling short, such as high rates of complications in certain patient populations.
- ✓ Gather feedback from patients, surgeons, and other members of the care team to understand their experiences with the program and identify areas for improvement.
- ✓ Analyze the data and feedback to identify trends and patterns, and use this information to inform adjustments and improvements to the program.
- ✓ Implement these changes and monitor their effectiveness through continued data collection and analysis.
- ✓ Repeat the process of data collection, analysis, and adjustment on a regular basis to ensure that the program remains effective and efficient over time.

Activity: Questions and Answers

Time: 20 minutes

Instruction

- ✓ Ask the questions to the whole group.
- ✓ Encourage active participation.
- ✓ Allow students to reflect/answer on implementation process of ERAS.
- ✓ Validate and summarize the answers provided.

Summary

Successful ERAS implementation depends on many factors, including patient and staff education, engagement, collaborative leadership culture between the facility and interdisciplinary team, and use of outcome data to continuously improve elements of the pathway. Steps of ERAS implementations are; **Conduc**t a thorough assessment of current practices and identify areas for improvement, **Develop** a multidisciplinary team to plan and implement the program, **Clear** protocols and guidelines for patient care in

enhanced recovery after surgery (ERAS), **Implement** evidence-based practices to optimize patient outcomes Incorporate, **Monitor** progress and continuously evaluate the program's effectiveness, **Provide** ongoing education and training for all team members, **Encourage** patient and family involvement in the program and **Use data and feedback** to make ongoing adjustments and improvements to the program.

Session 2.3. Possible Barriers of Implementing ERAS

Session Duration; 60 min

Session objectives: At the end of this session participants will be able to:

- ✓ Explain possible barriers to implement ERAS programs

2.3.1. Introduction

Activity: Small group discussion and gallery walk

Time: 15 minutes

Instruction

- ✓ List and describe ERAS implementation barriers
- ✓ Write the lists with flip-chart
- ✓ Each team present the discussed topic
- ✓ Summarize, validate and give feedback on the presented topics

Despite the potential benefits of ERAS, there are still many barriers to its implementation. These include lack of knowledge or training among healthcare professionals, lack of resources and funding, organizational challenges such as a lack of consistent protocol and inadequate communication between stakeholders, and patient-related issues such as non-adherence and lack of understanding of the importance of pre-operative preparation. Additionally, there may be cultural or societal beliefs that can lead to a decrease in compliance with the recommended procedures.

2.3.2 Common barriers to implement ERAS

Lack of understanding and knowledge about ERAS among healthcare professionals: Many surgeons, nurses, and other healthcare providers may not be familiar with the principles and practices of ERAS. It is important to continue educating and informing healthcare professionals on the benefits of enhanced recovery

after surgery and how to incorporate it into their practice. Additionally, research and publications on this topic should be provided to healthcare professionals so that they are aware of the developments in this field.

Resistance to change: Implementing ERAS protocols may require significant changes to existing surgical practices and routines, which can be met with resistance from healthcare professionals who are comfortable with the status quo. To address this issue, it is important to consider why resistance to change exists. Common reasons may include lack of knowledge and understanding of the new process, or difficulty in adapting to new ways of working. Once you have identified the causes of resistance, you can then begin to create strategies to address them.

Financial constraints: Implementing ERAS protocols may require additional resources and funding, which can be a barrier for healthcare organizations with limited budgets. First off, it's important to identify what health facilities budget is and look for cost-effective options that meet health facilities financial constraints. Additionally, there are several ways to reduce the cost of implementation, such as utilizing already available resources and technologies, negotiating for cost savings with suppliers, and strategizing about ways to reduce labor costs. Finally, health facilities may want to consider seeking out grants or donations to help offset any costs.

Limited standardization: There is currently a lack of standardization in ERAS protocols, making it difficult for healthcare organizations to adopt a consistent approach to ERAS. Enhanced recovery after surgery (ERAS) protocols should be tailored to the individual patient and the specific procedure. Standardization should be limited to the elements that have been proven to be effective and safe. These elements should include preoperative patient education, preoperative optimization, intraoperative techniques, and postoperative care. The implementation of ERAS protocols should be monitored and evaluated to ensure that the desired outcomes are achieved.

Lack of evidence-based medicine: There is still a lack of evidence-based medicine to support the use of ERAS protocols. Lack of interdisciplinary approach: ERAS protocol requires interdisciplinary approach which is not always possible to achieve in all clinical settings.

Activity: Questions and Answers

ERAS implementation barriers

Time: 20min

Instruction

- ✓ Ask the questions to the whole group
- ✓ Encourage active participation
- ✓ Allow students to reflect/answer on the purpose of clinical audit
- ✓ Validate and summarize the answers provided

Summary

Despite the potential benefits of ERAS, there are still many barriers to its implementation. These include lack of knowledge or training among healthcare professionals, cultural or societal beliefs that can lead to a decrease in compliance with the recommended procedures and Financial constraints and lack of evidence-based medicine.in order to solve these problems Health facilities may want to consider seeking out grants or donations to help offset any costs, continue educating and informing healthcare professionals on the benefits of enhanced recovery after surgery ERAS and protocols should be tailored to the individual patient and the specific procedure.

Session 2.4. Improving Multidisciplinary Professional Communication in ERAS

Session duration: 60 Minutes

Session Objectives: At the end of this session participants will be able to:

- ✓ Identify ways of effective professional communication
- ✓ Describe Multidisciplinary knowledge transfer
- ✓ Identify ways of improving Multidisciplinary professional communication

2.4.1 Introduction

Activity: Small group discussion and reflection

1. What is multidisciplinary professional communication?

Time: 15 minutes

The ERAS pathway is carried out in the form of multidisciplinary cooperation, including multiple modes of surgery, anaesthesia, nursing and perioperative management to ensure the safety and effectiveness of patients during the perioperative period. Overall, the establishment of a patient-centred multidisciplinary team (MDT) is an inevitable trend in the development of ERAS. However, the construction and management of ERAS–MDT is still in the exploratory stage, and multidisciplinary members still face many barriers and challenges in the various processes of implementing ERAS.

2.4.2. Effective professional communication

Effective communication among staff encourages effective teamwork and promotes continuity and clarity within the patient care team. At its best, good communication encourages collaboration, fosters teamwork, and helps prevent errors. ERAS elements are carried out by various professional departments, such as surgery, anesthesia, nursing, and physical therapy. The sharing of patients' information and communication between ERAS participants are key to successful multidisciplinary collaboration. Formal structures to encourage communication, such as consistent ERAS team meetings, can be helpful for multidisciplinary collaboration.

2.4.3. Multidisciplinary knowledge transfer

The success of ERAS depends on the effective transfer of knowledge between the various disciplines involved. This includes the sharing of information between the surgeon, anesthesiologist, nurses, physical therapists, dietitians, and other healthcare professionals. This knowledge transfer is essential for ensuring that the patient receives the best possible care and that the patient's recovery is as successful as possible.

The knowledge transfer between the various disciplines involved in ERAS is facilitated by the use of evidence-based protocols and guidelines. These protocols and guidelines provide a framework for the various disciplines to work together and ensure that the patient receives the best possible care. Additionally, the use of technology, such as electronic medical records, can help facilitate the knowledge transfer between the various disciplines.

In conclusion, ERAS is a multidisciplinary approach to patient care that involves the effective transfer of knowledge between the various disciplines involved. This knowledge transfer is essential for ensuring that the patient receives the best possible care and that the patient's recovery is as successful as possible. The use of evidence-based protocols and guidelines, as well as the use of technology, can help facilitate the knowledge transfer between the various disciplines.

2.4.4. Improving multidisciplinary professional communication

To improve multidisciplinary professional communication in ERAS, it is important to ensure that all members of the care team are aware of their roles and responsibilities, as well as the roles and responsibilities of other care team members. Additionally, it is important to create a framework for communication between members of the care team, such as regular meetings and open lines of communication. It is also important to ensure that all members of the care team have access to the same information, such as patient records and test results. Finally, it is important to encourage collaboration between members of the care team to ensure that the best possible outcomes are achieved.

Activity: Questions and Answers

1. What are components of Multidisciplinary professional communication?

2. Describe characteristics of effective multidisciplinary professional communication?

3. List advantages of effective multidisciplinary professional communication

Time: 25 Minutes

Summary

Enhanced recovery after surgery (ERAS) is a multidisciplinary approach to patient care that involves collaboration between various healthcare professionals, including surgeons, anesthesiologists, nurses, physical therapists, dietitians, and pharmacists. Communication among these professionals is essential for the successful implementation of ERAS protocols, as it allows for the sharing of information and the development of a comprehensive treatment plan. Effective communication also helps to ensure that all members of the care team are on the same page and that the patient's needs are being met. Additionally, communication helps to reduce the risk of errors and improves patient safety.

Session 2.5. Quality assurance steps in ERAS

Session duration: 60 minutes

Session Objectives: At the end of this session participants will be able to:

- Describe quality assurance steps
- Recognize tools used in quality assurance of ERAS programs

2.5.1. Introduction

Activity: Small group discussion and presentation

Discuss quality assurance steps and tools used in quality assurance of ERAS

Time: 20 min

Instruction;

Be in two groups and discuss

- Group 1: quality assurance steps
- Group 2: discus on tools used in quality assurance of ERAS program

2.5.2. Steps of quality assurance in ERAS

Quality assurance measures in health systems involve a set of activities and processes that are put in place to ensure that the healthcare services provided meet certain standards of quality and safety. These measures can include things like regular audits and inspections, performance monitoring, and the use of evidence-based guidelines and protocols. Some specific examples of quality assurance measures in health systems include:

Accreditation: Health systems may seek accreditation from organizations such as Ethiopian national accreditation office (ENAO) to demonstrate that they meet certain standards of quality and safety.

Clinical practice guidelines: Many health systems use clinical practice guidelines, which are evidence-based recommendations for the diagnosis and treatment of various medical conditions, to ensure that patients receive consistent and appropriate care.

Performance monitoring: Health systems may track and report on a variety of performance indicators, such as patient satisfaction, infection rates, and readmission rates, to identify areas where improvements are needed.

Patient safety: Many health systems have implemented patient safety programs to reduce the risk of errors and adverse events, such as medication errors and infections.

Specifically implementing quality assurance measures in health systems can help to improve the quality of care provided to patients, and also can be a way to reduce costs in the long term.

The health institute can assure the quality-of-service provision using Enhanced Recovery After Surgery (ERAS) by implementing the following steps:

- ✓ Develop and implement evidence-based protocols for ERAS that are tailored to specific surgical procedures.
- ✓ Provide education and training for all members of the healthcare team on the ERAS protocols and best practices.
- ✓ Continuously monitor and evaluate the outcomes of patients who have undergone surgery using ERAS protocols, and use this data to make adjustments and improvements to the protocols as needed.
- ✓ Foster a culture of continuous improvement and collaboration among all members of the healthcare team, including surgeons, nurses, and anesthesiologists.
- ✓ Ensure that appropriate patient selection and preparation are implemented before surgery.
- ✓ Incorporate patient involvement and communication throughout the process to ensure they understand the plan and have realistic expectations.
- ✓ Establish a multidisciplinary team (anesthesia, nursing, surgery) to monitor the ERAS process and make adjustments as needed.
- ✓ Continuously evaluate and update the protocols to ensure they are up-to-date with the latest research and evidence.
- ✓ Implement quality assurance and quality control measures to ensure that the ERAS protocols are being followed correctly and that the desired outcomes are being achieved.

- ✓ Report the ERAS outcomes to the public and professional organizations.

2.5.3 Tools used in quality assurance of ERAS

There are several tools and methods used for quality assurance (QA) in ERAS protocols. These include:

- ✓ Audits: Regularly reviewing patient outcomes and comparing them to established benchmarks and guidelines.
- ✓ Data collection and analysis: Tracking patient outcomes using standardized data collection forms and analyzing the data to identify areas for improvement.
- ✓ Clinical pathways: Using standardized protocols for patient care and monitoring adherence to these protocols.
- ✓ Multidisciplinary teams: Involving a team of healthcare professionals, including surgeons, nurses, anesthetists, and physical therapists, in the development and implementation of ERAS protocols.
- ✓ Patient and family engagement: Involving patients and their families in the QA process to gather feedback and identify areas for improvement.
- ✓ Feedback and continuous improvement: Continuously monitoring, evaluating and making adjustments to the ERAS protocol based on patient outcomes and feedback.
- ✓ Benchmarking: Comparing the outcomes of the ERAS protocol to those of other hospitals or programs to identify areas for improvement.

Activity: Questions and Answers

Quality assurance steps and tools used in quality assurance of ERAS

Time: 20min

Instruction to conduct Question and Answer

- ✓ Ask the questions to the whole group
- ✓ Encourage active participation
- ✓ Allow students to reflect/answer on the purpose of clinical audit
- ✓ Validate and summarize the answers provided

Summary

- Quality assurance measures are put in place to ensure that healthcare services provided meet certain standards of quality and safety.
- These measures can include things like regular audits and inspections, as well as the use of evidence-based guidelines and protocols.
- Some examples of quality assurance measures in health systems include accreditation and patient safety programs.
- There are several tools and methods used for quality assurance (QA) in enhanced recovery after surgery (ERAS) these include auditing, data collection and analysis.
- Quality assurance involves a team of healthcare professionals, including surgeons, nurses, anesthesiologists, and physical therapists.

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Chapter Three: The ERAS Bundles of Care

Duration: 720minutes

Chapter description: This Chapter focuses on ERAS care bundles that are evidence-based, multidisciplinary protocols that aim to optimize patient outcomes in the perioperative period.

Chapter Objective: At the end of this Chapter participants will be able to:

- Explain the ERAS care bundles recommended during perioperative period aiming for better patient outcome and reducing hospital stay and cost

Enabling Objectives: At the end of this Chapter participants will be able to

- Define ERAS care bundles
- Identify different ERAS care bundles recommended during preoperative, intraoperative and postoperative period
- Identify ways of preoperative pt education and optimization
- Identify how to provide and manage nutrition in ERAS
- Describe pain protocol and opioid pandemics
- Determine fluid management principles in ERAS
- Explain how to prevent postoperative nausea and Vomiting
- Identify ways of bowel preparation in ERAS protocol
- Elaborate how to prevent thromboembolism during perioperative period
- Explain anesthesia management principles in ERAS
- Identify ERAS principles and implementation in PACU/recovery room
- Identify and

Sessions
Session 3.1 Preoperative Patient Education and optimization
Session 3.2 Nutrition in ERAS
Session 3.3 Pain Management in ERAS and Opioid pandemics
Session 3.4 Fluid management in ERAS
Session 3.5 Postoperative nausea and vomiting (PONV) prevention
Session 3.6 Bowel preparation in ERAS
Session 3.7 Thromboembolism prevention
Session 3.8 General and regional anesthesia management in ERAS
Session 3.9 ERAS in the PACU/recovery room
Session 3.10 Epidural and caudal blocks
Session 3.11 Chest blocks (including erector spinae block)
Session 3.12 Abdominal wall blocks

Session 3.1 Patient education and multidisciplinary optimization in ERAS

Duration 120 Minutes

Session objectives

- Describe components of patient education
- Identify concepts of effective communication
- List mechanism of Perioperative risk analysis in an Enhanced Recovery after Surgery
- Describe concepts of Informed consent
- List Multidisciplinary optimization of common comorbidities
- List Impacts and management of preoperative anxiety

Activity: Gallery walk

Provide six topic for six group to prepare a presentation for gallery walk

Topics 1,patient education 2,effective communication 3,peropreative risk analysis,4,informed consent form 5, Multidisciplinary optimization of common comorbidities 6, Impacts and management of preoperative anxiety

Duration: 30 min

Instruction to conduct Gallery Walk

Provide the assignment 30 min before the session

Prepare the space

Prepare the materials

Introduce the activity

Divide into six groups

Allow time for discussion

Wrap up

Patient education is an important component of ERAS, as it helps patients understand the expectations and requirements of their surgery and recovery.

The goal of pre-operative patient education in ERAS is to empower patients with knowledge and skills to optimize their recovery, reduce the risk of complications, and improve overall outcomes.

3.1.1 Introduction

Pre-operative patient education is an important component of ERAS which helps the patient to be informed about the pre-operative fasting, medications, and other preparation requirements. Pre-operative care covers the topics that patients must understand in order to receive adequate care these are;

- Explanation of the procedure and expected outcomes
- Discussion of pre-operative fasting and fluid guidelines
- Review of medications and any necessary adjustments
- Instructions for physical activity and mobility
- Discussion of potential side effects and post-operative care
- Encouragement of active participation in the recovery process.

Patient education on Anesthesia and pain management: Patients should be educated about the types of anesthesia and pain management options available during and after surgery.

- Explain the purpose of anesthesia during surgery: Anesthesia helps to control pain and discomfort during a surgical procedure.
- Discuss the different types of anesthesia: General, regional, and local anesthesia.
- Explain the role of pain management which aims to minimize pain and discomfort after surgery to promote faster recovery.
- Discuss on Non-steroidal anti-inflammatory drugs (NSAIDs), acetaminophen, nerve blocks, and other non-opioid pain management options.
- Explain the importance of Proper pain management after surgery which helps to reduce the risk of complications, such as deep vein thrombosis, pneumonia, and wound infections.
- Discuss the potential side effects of pain medication, like nausea, dizziness, constipation, and other side effects that can occur with pain medication.
- Emphasize the importance of patient communication with their healthcare provider about their level of pain and any side effects they are experiencing.
- Encourage patients to take an active role in their pain management by using relaxation techniques, such as deep breathing and visualization, and by moving around as much as possible after surgery.

Early mobilization: Patients should be encouraged to start moving as soon as possible after surgery to promote healing and reduce the risk of complications. Early mobilization after surgery is a key component of Enhanced Recovery After Surgery (ERAS) programs. The goal of early mobilization is to get patients up and moving as soon as possible after surgery to reduce the risk of complications, such as blood clots, pneumonia, and muscle weakness. Early mobilization also helps with pain management and improves patients' overall well-being and recovery. It is typically started within 24 hours of surgery and gradually increased as the patient's condition improves.

Nutrition: Patients should be advised on the importance of good nutrition and hydration during the recovery period. Importance of good nutrition before and after surgery: Good nutrition helps the body heal faster and reduces the risk of complications.

- ✓ Protein: Consuming adequate amounts of protein can help repair and rebuild tissues damaged during surgery.
- ✓ Hydration: Staying hydrated is important for proper wound healing and reducing the risk of infection.
- ✓ Fiber: Fiber helps prevent constipation, a common side effect of pain medications and inactivity.
- ✓ Vitamins and minerals: Vitamins and minerals, such as iron and vitamin C, are important for wound healing and overall health.
- ✓ Avoiding unhealthy foods: High-fat, high-sugar, and processed foods can slow down recovery and increase the risk of complications.
- ✓ Gradual return to normal diet: Gradually increasing physical activity and returning to a normal diet can help prevent complications and improve recovery.
- ✓ Consult with a dietitian: A dietitian can provide personalized nutrition recommendations and help ensure that the patient is getting the nutrients they need for optimal recovery.

Medications: Patients should be informed about the medications they will need to take during their recovery and their possible side effects. Taking medications as prescribed is an important part of the program and can help reduce pain, prevent infection, and speed up recovery.

Wound care: Patients should be taught how to care for their incision site and how to recognize signs of infection. Some of patent education we should give are

- ✓ Clean the wound daily with saline solution or soap and water.
- ✓ Keep the wound covered with a sterile dressing.
- ✓ Avoid soaking the wound in water.
- ✓ Report any signs of infection, such as increased pain, redness, or swelling.
- ✓ Keep the wound elevated to reduce swelling.
- ✓ Follow a balanced diet to promote healing.

- ✓ Avoid smoking, as it can slow down the healing process.
- ✓ Follow any additional instructions from your healthcare provider.

Follow-up care: Patients should be informed about the follow-up appointments and tests they will need after their surgery. Lifestyle changes: Patients should be encouraged to make healthy lifestyle changes, such as quitting smoking, to improve their recovery and overall health.

3.1.2 Effective communication

Communication between the patient and healthcare provider is key to a successful recovery after surgery. Good communication can help reduce anxiety and stress levels, which can have a positive impact on the patient's recovery. It can also improve patient satisfaction, as they feel heard and well-informed about their care. Some of the importance of Effective communications is,

- ✓ Clarity of expectations: Clear communication between the patient and healthcare provider can ensure that the patient has a clear understanding of what to expect during the recovery process.
- ✓ Reduced anxiety: Good communication can help reduce anxiety and stress levels, which can have a positive impact on the patient's recovery.
- ✓ Improved adherence to care plan: Effective communication can help ensure that the patient follows the recommended care plan, which can speed up the recovery process.
- ✓ Early identification of complications: Prompt and clear communication can help identify any complications early on, allowing for timely intervention and improved outcomes.
- ✓ Better patient satisfaction: Good communication can lead to increased patient satisfaction, as the patient feels heard and well-informed about their care.

3.1.3 Perioperative risk analysis in an Enhanced Recovery after Surgery

Perioperative risk analysis involves evaluating the patient's risk factors prior to surgery to help guide decision making and improve outcomes. The goal of perioperative risk analysis is to identify any potential risk factors and develop a plan to mitigate these risks, leading to improved surgical outcomes and a faster recovery. These can include:

1. Medical history review: This includes evaluating the patient's past medical history for conditions such as heart disease, diabetes, and other chronic illnesses that may increase the risk of complications during and after surgery.

2. Physical examination: This includes evaluating the patient's current physical status and any signs of poor health, such as malnutrition or dehydration.

3. Laboratory tests: This includes blood work and other tests to assess the patient's overall health and identify any underlying conditions that may increase the risk of complications.

4. Imaging studies: This includes X-rays, CT scans, and other imaging tests to evaluate the patient's anatomy and identify any potential issues that may impact the surgical outcome.

Discussing perioperative risk analysis with a patient

Is important for several reasons:

- ✓ Patient education: It helps patients understand the potential risks and benefits of their surgery, allowing them to make informed decisions.
- ✓ Improved outcomes: By identifying and addressing potential risks, the patient and healthcare team can work together to minimize them and improve postoperative outcomes.
- ✓ Better communication: The discussion helps establish clear expectations and open lines of communication between the patient and healthcare team.
- ✓ Increased patient involvement: Patients who are involved in the decision-making process are more likely to adhere to postoperative care plans, leading to better outcomes.
- ✓ Improved patient satisfaction: By involving patients in the risk analysis process, they feel more involved in their care and are more likely to be satisfied with their surgical experience.

Informed consent

The first step in obtaining informed consent is to provide the patient with a clear and comprehensive explanation of the procedure, including its purpose, steps involved, and potential risks and benefits. This helps the patient understand what they can expect and make an informed decision about whether to proceed with the surgery.

The healthcare provider should also discuss alternative treatments and procedures with the patient, and explain why enhanced recovery after surgery may be the best option. This helps the patient understand why the procedure is being recommended and what other options are available.

Before the procedure, the patient should be given an informed consent form to review and sign. This form should include all of the information discussed during the informed consent process, including the procedure,

risks, benefits, and alternatives. The healthcare provider should take the time to review the form with the patient, answer any questions, and ensure that the patient fully understands the information before signing.

3.1.4 Multidisciplinary optimization of common comorbidities

Enhanced recovery after surgery (ERAS) is a protocol that aims to optimize patient outcomes through a multidisciplinary approach to preoperative, intraoperative, and postoperative care. The optimization of common comorbidities, such as obesity, diabetes, and cardiovascular disease, is a key component of ERAS. This involves addressing these conditions through preoperative assessment, perioperative management, and postoperative rehabilitation. The goal is to reduce the risk of complications and improve patient outcomes, including shorter hospital stays, reduced pain, and improved quality of life

Obesity; It is important to take into account the patient's physical health and health goals when addressing obesity during perioperative care. The most effective solutions are tailored to each individual patient. Interventions could include physical activity, lifestyle modifications such as calorie reduction, cognitive behavioural therapy (CBT), and nutrition counseling.

Diabetes Optimization of diabetes management is an important part of ERAS programs. Strategies include preoperative optimization of glycemic control, intraoperative management of glucose levels, and postoperative management of diabetes medications. Additionally, patient education is an important part of ERAS programs to ensure that patients are aware of their diabetes and how to manage it during their recovery.

Cardiovascular disease can be optimized through the following strategies:

- Optimizing preoperative patient assessment and optimization of risk factors for cardiovascular diseases.
- Implementing an individualized nutrition program during the perioperative period.
- Utilizing pharmacologic interventions to reduce cardiovascular risk.
- Implementing an early mobilization program to improve cardiovascular outcomes.
- Utilizing preoperative cardiac stress tests to identify patients at risk for cardiovascular events.
- Implementing a postoperative surveillance program to identify patients at risk for cardiovascular events.
- Utilizing tele monitoring to reduce hospital readmissions and improve cardiovascular outcomes.
- Implementing an evidence-based approach to the management of postoperative pain.
- Implementing strategies to reduce the risk of venous thromboembolism.
- Utilizing preoperative and postoperative monitoring of blood pressure, heart rate, and other cardiovascular parameters.

3.1.5 Impacts and management of preoperative anxiety

Preoperative anxiety can have a significant impact on patient outcomes, including increased pain, Anxiety can trigger the body's fight-or-flight response, which can cause increased tension and pain in the body. Additionally, anxiety can cause an increase in stress hormones which can lead to further pain

Increased risk of postoperative complications, Anxiety has been shown to have a significant impact on postoperative complications increased risk of infection, increased risk of wound healing complications, increased risk of readmission, and increased risk of mortality and delayed recovery. To manage preoperative anxiety, healthcare providers may use a variety of techniques, including patient education, relaxation techniques, distraction, and pharmacological interventions. Additionally, providing emotional support and a positive environment can help reduce preoperative anxiety.

Activity: Question and Answer

List the purpose of clinical audit template

Duration: 20 min

Instruction to conduct Question and Answer

- Ask the questions to the whole group
- Encourage active participation
- Allow students to reflect/answer on the purpose of clinical audit
- Validate and summarize the answers provided

Session summary

The goal of pre-operative patient education in ERAS is to empower patients with knowledge and skills to optimize their recovery, reduce the risk of complications, and improve overall outcomes. Pre-operative care covers the topics that patients must understand in order to receive adequate care. The goal of early mobilization is to get patients up and moving as soon as possible after surgery to reduce the risk of complications, such as blood clots, pneumonia, and muscle weakness. Early mobilization also helps with pain management and improves patients' overall well-being and recovery.

Communication between the patient and healthcare provider is key to a successful recovery after surgery. Good communication can help reduce anxiety and stress levels, which can have a positive

impact on the patient's recovery. It can also improve patient satisfaction, as they feel heard and well-informed about their care.

The goal of perioperative risk analysis is to identify any potential risk factors and develop a plan to mitigate these risks. This can include evaluating the patient's medical history for conditions such as heart disease, diabetes, and other chronic illnesses that may increase the risk of complications during and after surgery.

The healthcare provider should provide a clear and comprehensive explanation of the procedure, including its purpose, steps involved, and potential risks and benefits. The patient should also be given an informed consent form to review and sign. This form should include all of the information discussed during the informed consent process, including the risks, benefits, and alternatives.

The optimization of common comorbidities, such as obesity, diabetes, and cardiovascular disease, is a key component of ERAS. This involves addressing these conditions through preoperative assessment, perioperative management, and postoperative rehabilitation. The goal is to reduce the risk of complications and improve patient outcomes.

Preoperative anxiety can have a significant impact on patient outcomes, including increased pain and mortality. To manage preoperative anxiety, healthcare providers may use a variety of techniques, including patient education, relaxation techniques, distraction, and pharmacological interventions. Additionally, providing emotional support and a positive environment can help reduce preoperative anxiety.

Session 3.2 Nutrition in ERAS

Duration 60 minutes

Session objective; at the end of this session participants will be able to: Identify the impact of nutritional status on surgical outcome, preoperative nutritional assessment and nutritional supplement and optimization

3.2.1 Introduction

Activity: group discussion **What is enhanced recovery after surgery in nutrition?** **Time: 10 minutes**

ERAS (Enhanced Recovery After Surgery) is a concept in surgical care that aims to optimize patient outcomes by reducing stress, pain, and complications during the postoperative period. In the context of nutrition, ERAS protocols often include guidelines for preoperative fasting, perioperative fluid and electrolyte management, and early oral or enteral nutrition. The goal is to promote recovery and reduce the length of hospital stay.

ERAS identifies nutrition status as a modifier of surgical outcomes, so nutrition-related interventions are integrated into several elements, and many of the elements directly or indirectly affect nutrition status. [1] In fact, implementation of the ERAS protocol has been found to transform surgical nutrition practices.2A prospective study of a provincial ERAS initiative for elective colorectal surgery in Canada identified that with an overall compliance rate to the ERAS elements of 72%, nutrition risk screening increased from 9% to 74%, postoperative feeding took place significantly earlier, and the number of days to tolerate solid food was reduced from 5.9 to 2.3 days.[2]

3.2.2 Screen for nutrition risk before surgery

The ERAS (Enhanced Recovery After Surgery) care component "Screen for Nutrition Risk before Surgery" refers to the process of evaluating a patient's nutrition status before undergoing surgery. This helps determine if the patient is at risk of malnutrition and if they require additional support or interventions to optimize their nutritional status before and after surgery. This step is important in the ERAS protocol, as optimal nutrition is essential for a successful recovery and can improve outcomes such as reduced length of hospital stay and improved wound healing.

Although there is no broadly accepted definition of malnutrition, it is often defined as an unbalanced nutrition state that leads to altered body composition and diminished function. It has been estimated that 24%–65% patients in developed countries present for elective surgeries in a state of malnutrition risk.

A multicentre prospective cohort study conducted across Canada in 2016 found that 45% of adult patients admitted to medical and surgical units were suffering from some degree of malnutrition.14 These malnourished patients stayed in hospital up to 3 days longer than their well-nourished counterparts, which was estimated to cost the healthcare system billions annually.14,15

Correction of perioperative malnutrition improves patient outcomes by decreasing postoperative complications19 and length of stay.20–22 As a result, the American Society for Parenteral Enteral Nutrition (ASPEN), European Society for Clinical Nutrition and Metabolism (ESPEN), the Academy of Nutrition and Dietetics, and the Canadian Malnutrition Task Force all recommend early identification and treatment of malnutrition.9

3.2.3 Avoid fasting with oral carbohydrate loading

The purpose of preoperative carbohydrate loading is threefold: (1) change pre-surgical metabolism from a fasted to a fed state, (2) attenuate postoperative insulin resistance, and (3) keep the patient hydrated.38 Perioperative fasting can exacerbate the surgical stress response, which enhances body protein losses.39,40 Under traditional surgical care models, a state of insulin resistance typically develops in otherwise healthy individuals and persists for 2–3 weeks even after uncomplicated, moderate, elective surgery.41 The consumption of carbohydrate drinks before surgery can induce a metabolically fed state by producing an insulin response that is similar to that experienced after ingestion of a meal.38 This metabolic shift inhibits the breakdown of energy/nutrient reserves (ie, inhibits catabolism), blunts amino acid oxidation (ie, use of body protein for fuel), improves patient comfort by minimizing dehydration and thirst, and modulates the surgical stress response by supporting a state of postoperative insulin sensitivity.38,41,42 Hyperinsulinemic euglycemic clamp studies (the gold standard for measuring insulin sensitivity) have demonstrated that preoperative carbohydrate loading improves postoperative insulin sensitivity.43,44 As an example, in a randomized, double-blinded, placebo-controlled study, oral carbohydrate loading with a 50-g, 12.5% hypo-osmolar carbohydrate beverage (12% monosaccharaides, 12% disaccharides, 76% polysaccharides; consumed within 10 min on the morning of total hip replacement surgery and approximately 2 h before the estimated time of insertion of the epidural catheter) decreased whole-body insulin sensitivity on the first postoperative day (POD) by 18% vs 43% in the placebo group.44

The concept of carbohydrate loading may be new to patients, and so patients may need numerous reinforcing messages. The dietitians can support this message by promoting oral carbohydrate loading in preoperative education sessions with patients and staff as well as by participating in the development of educational materials. Additionally, to ensure the carbohydrate-loading processes have been done correctly, we recommend use of a surgery checklist that includes asking whether a carbohydrate beverage was taken, along with the type and timing of the beverage consumed. Considering patients are expected to arrive at the hospital 2–3 h prior to their operation, acceptable beverages can be stocked at the preadmission clinic.

3.2.4 Begin oral nutrition early postoperatively

Traditionally, institutions exercised slow resumption of regular oral food intake (ie, began with fluids, then soft foods) upon the return of bowel function.74 The rationale behind this traditional approach was to "rest" the bowel; however, physiological studies have identified that the bowel continues to be active, producing large amounts of digestive fluids, even in the fasted state.74,75 Moreover, clinical and physiologic studies of gastrointestinal motility after surgery have found that small-bowel motility resumes within hours of the operation,76,77 and over 90% of digestion and absorption occurs in the small intestine.74 Early feeding refers to providing food and oral nutrition supplements to patients within hours after their operation. Several meta-analyses have demonstrated the safety and efficacy of oral intake within 24 h after abdominal surgery.78,79 After colorectal surgery, early feeding is associated with positive clinical outcomes such as reduced mortality, anastomotic dehiscence, and length of hospital stay.80–83 Additionally, early feeding may stimulate gut hormones and propulsive activity,84 which encourages normal resumption of bowel activity and tolerance of oral intake.85 It is important to note, however, that early feeding might not be tolerated if the previous ERAS elements are not followed closely.

Activity: group discussion

1. Describe types of foods recommended in ERAS
2. Elaborate dietitians involvement for nutrition in ERAS
3. Describe principles of nutritional support in ERAS

   Time : 30 minutes

3.2.5 Summary

For ERAS to truly be effective in enhancing patient experiences and improving outcomes, all elements need to be implemented, and this requires a collaborative team approach. The dietitians can be an active team member in the implementation and sustainment of an ERAS program. Nutrition is a key component of many ERAS care elements. Nutrition screening before surgery can allow for early detection of malnutrition and treatment. Avoiding fasting with oral carbohydrate loading attenuates post-surgical insulin resistance, enhances postoperative utilization of glucose, and spares body protein. Proactive management of PONV supports food intake and improves the patient experience. Judicious fluid management prevents gastrointestinal dysfunction and patient discomfort, encouraging tolerance of early oral feeding. Early oral feeding and ONS enable patients to meet nutrition needs, maintain a neutral fluid balance/weight status, and return home sooner. Finally, auditing permits assessment of the protocol in practice, which allows for needed changes to patient care and acknowledgment of successes, enabling better patient care, experiences, and outcomes. The dietitian plays an important role in carrying out these elements and thus is integral to the success of an ERAS program.

Session 3.3: Pain Management in ERAS and Opioid pandemics

Session duration: 60 Minutes

Session objective: At the end of this chapter, participants will be able to

- ✓ Describe the consequences of untreated acute perioperative pain
- ✓ Use different pain assessment tools
- ✓ Discuss appropriate modalities of pain management
- ✓ Explain opioid pandemics
- ✓ Recognize adverse effects of opioids in ERAS perspectives

3.3.1 Introduction

Activity: Individual Reflections

Why perioperative pain must be appropriately managed?

Time: 5 minutes

Perioperative pain control is important for humane reasons. The philosophy of patient-centred care considers pain management as one of the basic human rights. One of the major elements of an ERAS is provision of optimal perioperative pain management, since optimal pain management facilitates early postoperative feeding, ambulation and rehabilitation, which in turn reduce hospital length of stay. In contrast, higher pain scores are associated with higher perioperative complications. Inadequate postoperative pain relief may have long-term consequences such as increase in readmission rate and development of chronic pain.

Activity: Small group discussion and gallery walk

What are the consequences of uncontrolled acute pain?

Time: 10 minutes

The consequences of uncontrolled acute pain are;

- ✓ Hemodynamic instability,
- ✓ Respiratory morbidities,
- ✓ Emotional disturbances,
- ✓ Myocardial ischemia/infarction,
- ✓ Wound healing problems,

- ✓ Patient/care giver dissatisfaction,
- ✓ Chronic pain and
- ✓ Long-term disability

3.3.2. Pain assessment

Pain can be assessed by using clinical parameters and tools.

Activity: Small group discussion and presentation
Discuss different pain assessment methods (including subjective and objective).
Time: 30 minutes

3.3.3 Pain management modalities in ERAS

The choice of analgesia will depend on the surgical approach, the position and sizes of incisions and patient factors. There also needs to be a plan for breakthrough pain and failure of the initial first-line technique. The rationale of the ERAS approach to analgesia is to use multimodal analgesia combined with regional and local anesthetic techniques to allow a patient's opiate consumption to be minimized. This avoids the unwanted side effects of sedation, nausea and vomiting, delirium and gut dysfunction.

There are few specific studies on analgesia within enhanced recovery after surgery pathways; so much of the evidence is extrapolated.

3.3.3.1 Multimodal non-opioid analgesia

Activity: Small group discussion and reflections
What is multimodal analgesia mean?
What are the advantages of multimodal analgesia within ERAS program?
Time: 5 minutes

The backbone of multimodal analgesia within an ERAS program is a major opioid-sparing local anesthetic technique such as use of regional blocks in different approaches such as catheter-based techniques combined with non-opioid medication. Acetaminophen (paracetamol) combined with a non-steroidal anti-inflammatory drug (NSAID) is a good combination. Opioids are best reserved for breakthrough pain. Other analgesic adjuncts such as gabapentinoids, magnesium, ketamine and lidocaine are being increasingly used.

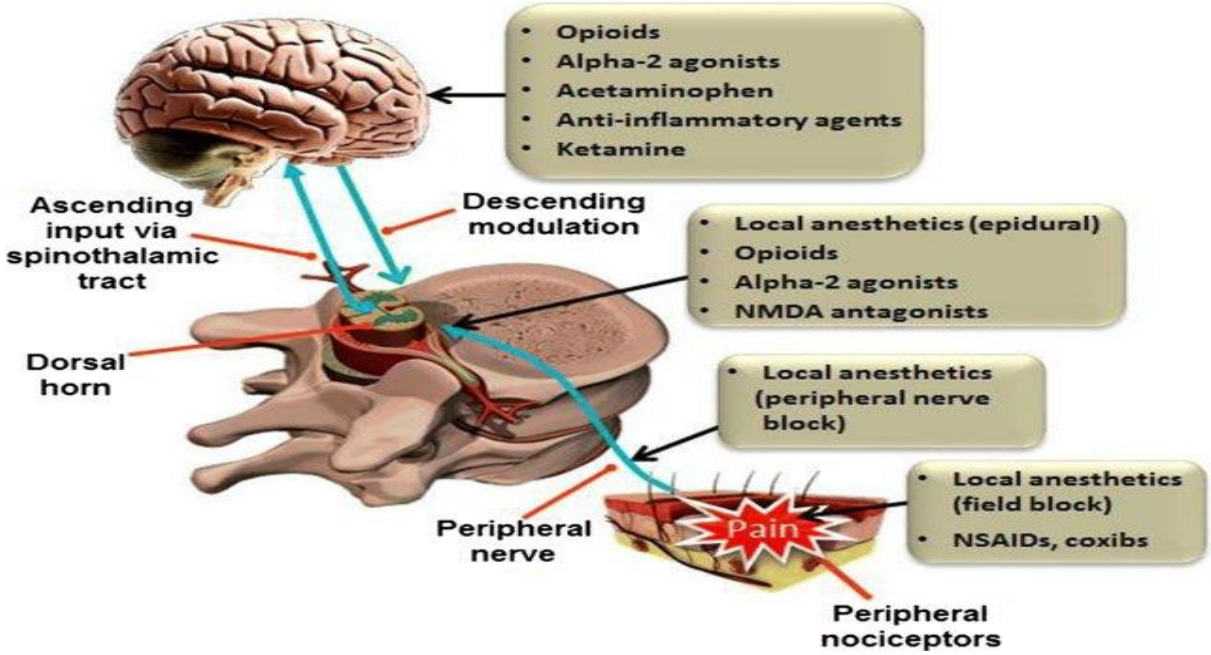

Figure 1: Multimodal approach to perioperative pain management effects on different phases of the pain pathway

Figure 2: Inversed pyramid of multimodal perioperative pain management

3.3.4. Opioid pandemics in ERAS perspectives

Activities: Small group discussion and reflections
Why clinicians should use opioid in healthcare settings appropriately?
How ERAS protocols can help to achieve appropriate use of opioids?
Time: 10 minutes

The opioid pandemic has had a devastating effect on many communities. As a result, there has been an increased focus on reducing opioid use and misuse in the healthcare setting. One way to do this is through the use of Enhanced Recovery After Surgery (ERAS) protocols. ERAS protocols are designed to reduce opioid use and misuse by providing evidence-based, multimodal pain management strategies that focus on minimizing opioid use. These strategies include the use of non-opioid analgesics, regional anesthesia, and multimodal analgesia. Additionally, ERAS protocols emphasize patient education, early mobilization, and early nutrition to reduce opioid use and improve patient outcomes.

The implementation of the ERAS protocols has been shown to reduce opioid use and misuse in the healthcare setting; as it can reduce opioid use by up to 50%, while also improving patient outcomes. Additionally, the ERAS protocols can reduce the risk of opioid-related adverse events, such as sedation, immobility, respiratory depression, nausea, and vomiting; in turn all of which have their own unwanted consequence. Furthermore, the ERAS protocols can help to reduce the risk of opioid addiction and overdose. Therefore, the ERAS protocols should be implemented in healthcare settings to help reduce the impact of the opioid pandemic.

3.3.5. Practical points in pain management in ERAS

- ✓ Enhanced recovery after surgery (ERAS) programs are increasingly becoming standard of care for several surgical procedures.
- ✓ Optimal pain management guidelines would be evidence-based, procedure-specific recommendations presented as preoperative, intraoperative, and postoperative interventions as well as surgical interventions that are easy to access, transparent, and relevant to clinicians.
- ✓ The aim of an optimal analgesic technique should be to facilitate early ambulation and physical therapy.
- ✓ Even if an analgesic technique provides excellent pain relief, it may not be clinically beneficial if the associated adverse events that delay recovery.
- ✓ Preoperative identification of patients at high risk of postoperative pain allows patient-specific approach to pain management and improves pain relief.
- ✓ Patients should be educated with regards to analgesic options and realistic expectations should be set.
- ✓ The aim an analgesic strategy should be to reduce pain intensity to an acceptable level that would improve functionality and allow ambulation, rather than achieve a certain pain score.
- ✓ Use of opioids (intraoperative and postoperative) should be reduced by using opioid-sparing analgesic approaches. However, the role of opioid-free anesthesia remains questionable.

- ✓ Combination of paracetamol and an NSAID or COX-2 specific inhibitor should be administered preoperatively or intra-operatively, and continued into the postoperative period, unless contraindicated.
- ✓ Use of gabapentinoids for surgical pain management remains controversial and there are concerns of potential adverse effects. Therefore, they should be used with great caution and at the lowest possible dose.
- ✓ Local/regional analgesia techniques should be a part of an optimal pain management strategy.
- ✓ Role of neuraxial blocks in the ERAS setting is decreasing due to concerns of adverse events that might delay ambulation and availability alternative analgesic techniques that can provide similar pain relief as well as similar recovery outcomes. Peripheral regional analgesia techniques such as fascial plane blocks and surgical site infiltration techniques are increasingly becoming popular due to their high success rate and minimal adverse effects.

Summary

- Multimodal analgesia involves use of multiple, simultaneous mechanisms of pain control acting synergistically to improve analgesic effect and reduce the doses of any single agent to minimize risks of side-effects.
- This aims to avoid opiates use altogether, or at least reduce opiate doses required, and thereby reduce risk of side effects from opiates that are common and cause delays in recovery.
- There are multiple pathways and mediators involved in nociception, and targeting several mechanisms can increase analgesic efficacy, using combinations of systemic and regional anesthesia.

Session 3.4 Fluid management in ERAS

Session duration: 50 Minutes

Session specific objectives:

- Identify Perioperative hydration assessment and monitoring
- Describe goal directed perioperative fluid therapy

Activity : Group discussion
What has been your experience in managing fluid in the ERAS?
10 minutes

3.4.1 Introduction

ERAS (Enhanced Recovery after Surgery) is a protocol aimed at improving patient outcomes and reducing recovery time after surgery. Fluid management is an important aspect of ERAS, as it helps to maintain fluid balance and prevent complications such as hypovolemia and electrolyte imbalances. The key principles of fluid management in ERAS include:

- Preoperative optimization: Patients are encouraged to drink clear fluids before surgery to maintain hydration.
- Intraoperative fluid management: Intraoperative fluid management is guided by continuous monitoring of hemodynamics and fluid balance, with a focus on avoiding over-resuscitation.
- Postoperative fluid management: After surgery, patients are encouraged to drink clear fluids and resume oral intake as soon as possible. IV fluids may also be administered to maintain fluid balance, but the amount and rate of fluid administration is carefully monitored to avoid over-resuscitation.
- Monitoring: Regular monitoring of fluid balance, electrolyte levels, and hemodynamics is crucial to ensure that fluid management goals are being met and to prevent complications.

3.4.2 Perioperative hydration assessment and monitoring

The goal of fluid management in ERAS is to maintain optimal fluid balance, prevent electrolyte imbalances, and avoid over-resuscitation. By following these principles, ERAS helps to promote faster recovery and improved patient outcomes.

ERAS (Enhanced Recovery after Surgery) protocols aim to optimize perioperative hydration by:

- Preoperative hydration: Providing clear fluids before surgery to prevent dehydration.
- Intraoperative fluid management: Administering balanced electrolyte solutions during surgery to maintain hydration and prevent fluid imbalances.
- Postoperative fluid management: Encouraging oral fluid intake and monitoring fluid balance to prevent dehydration and electrolyte imbalances.

- Monitoring: Regularly measuring fluid balance, electrolyte levels, and urine output to assess hydration status and adjust fluid management accordingly.
- Education: Educating patients on the importance of adequate hydration and providing guidelines for fluid intake before and after surgery.

3.4.3 Goal directed perioperative fluid therapy Goal-directed fluid therapy (GDFT), also known as hemodynamic-guided fluid therapy, is an important part of ERAS. It is focused on achieving the right balance of fluids and electrolytes to ensure optimum patient health and reduce the risk of complications. Studies are indicating that, this method is a superior deviation from the historical "fixed-volume" or liberal fluid approach. In a meta-analysis published by Chong et al, GDT was associated with lower risk of mortality, pneumonia, acute kidney injury, and wound infection as well as portended a shorter length of hospital stay. GDT appears to be most effective in the context of ERAS protocols, which implement multiple processes reducing fluid imbalances throughout the perioperative period, including avoidance of pre-operative dehydration and encouraging early post-operative discharge and ambulation. Overall aim of goal directed fluid therapy is to maintain optimum hemodynamic balance by avoiding both hypervolemia and hypovolemia (2). .

Activity: Group discussion

1. What are hemodynamic parameters to be monitored for goal directed fluid therapy in ERAS?
2. What monitoring standards are recommended for appropriate fluid balance in ERAS?

Time: 20 minutes

3.4.4 Session summary

The Enhanced Recovery After Surgery (ERAS) Society provides a comprehensive set of guidelines related to fluid management in perioperative care. These guidelines are evidence-based and serve to optimize outcomes of surgical patients. The guidelines cover a range of fluid management topics including fluid selection, fluid balance, fluid restriction, and fluid delivery. Additionally, the guidelines provide specific recommendations related to the timing of fluid administration and a monitoring schedule for perioperative fluid administration.

Session 3.5 Postoperative nausea and vomiting (PONV) prevention

Session duration: 40 minutes

Session objectives at the end of this session

Define basic pathophysiology of PONV

Improve Impacts of PONV

Assess the risk of having PONV

Increase Prevention and management of PONV

3.**5.1. Introduction**

Activity: Individual Reflections

How often post-operative nausea vomiting occur among surgical patients?

What is basic causes of PONV?

What are consequences of PONV?

Time: 5 minutes

Post-operative nausea and vomiting (PONV) are two of the most common and unpleasant side effects following anesthesia and surgery.53 PONV have been found to occur in approximately one-third of all surgery patients and up to 80% in patients at high risk for PONV.

The causes of PONV are multifactorial and include type and duration of surgery as well as use of inhalational anesthesia and opioids. Fluid overload is also a common factor causing delayed gastrointestinal motility, contributing to PONV.

PONV can delay discharge from the post-anesthesia care unit, increase risk of pulmonary aspiration, promote postoperative discomfort, delay early postoperative feeding, delay ONS intake, encourage bedrest, and prolong intravenous (IV) fluids and may require nasogastric tube placement if vomiting cannot be controlled—all of which delay discharge from hospital.

3.5.2 The risks and impacts of having PONV

Activity: Group discussion and gallery walk

What are the risk of post-operative nausea vomiting?

How to assess risk of PONV?

Time: 10 Minutes

To attenuate PONV, the ERAS protocol for surgery recommends use of opioid-sparing analgesia and adoption of a multimodal approach to PONV.In all patients with one or two risk factors for PONV, prophylactic medications should be considered. Risk factors for PONV include female sex, age age<50 years ,a history of PONV or motion sikness ,and smoker status

3.5.3 Prevention of PONV

Postoperative Nausea and Vomiting (PONV) prevention in Enhanced Recovery After Surgery (ERAS) protocols typically involves a combination of:

1. Patient selection and risk assessment
2. Multimodal antiemetic therapy (e.g. combination of drugs, nerve blocks)
3. Avoidance of fasting and provision of clear fluids before surgery
4. Optimal pain management
5. Minimizing use of opioids
6. Early mobilization and feeding

These interventions aim to reduce the risk factors that contribute to PONV and improve patient outcomes.

3.5.4 Summary and recommendation:

Activity Question and Answer

What is the best approach to prevent PONV?

Time : 5 minutes

A multimodal approach to PONV prophylaxis should be considered in all patients and incorporated into ERAS protocols. Patients with 1–2 risk factors should ideally receive a two-drug combination prophylaxis using first-line antiemetics. Patients with C 2 risk factors undergoing colorectal surgery should receive 2–3 antiemetics. If nausea and or vomiting still occur, despite prophylaxis, salvage therapy should be provided using a multimodal approach using different classes of drugs from those used for prophylaxis.

References

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2. Gustafsson UO, Scott MJ, Hubner M, et al. Guidelines for perioperative care in elective colorectal surgery: enhanced Recovery After Surgery (ERAS(R)) Society Recommendations: 2018. World J Surg. 2019;43(3):659-695.

Session 3.6 Bowel preparation in ERAS

Session duration 60 minutes

Session description this chapter deals with ERAS recommendation for bowl preparation

Session objectives at the end of this session participants will be able to Provide appropriate bowel preparation Implement ERAS recommendations during bowel preparation

3.6.1 Introduction

Activity group discussion Appropriate bowl preparation in ERAS

The goal of the preoperative phase of ERAS is for patients to obtain the energy necessary for the body to accommodate the high metabolic demands imposed by surgery. The traditional fasting requirements of surgery deplete liver glycogen and are associated with impaired glucose metabolism and increased insulin resistance, which have been shown to adversely affect perioperative outcomes. In contrast with traditional "nothing by mouth" strategies, ERAS pathways avoid dehydration by reducing the preoperative starvation period and utilizing complex carbohydrate drinks in nondiabetic patients. This strategy has been shown to reduce preoperative thirst and anxiety and reduce postoperative insulin resistance in colorectal surgery, ultimately reducing length of stay and improving patient satisfaction

3.6.2 Techniques of bowel preparation

Activity Gallery walk Instruction to conduct Gallery Walk ❖ Provide the assignment one hour before the session ❖ Prepare the space ❖ Prepare the materials ❖ Introduce the activity ❖ Divide into groups ❖ Allow time for discussion Wrap up Time: 30 minutes

ERAS Bowel Prep

This steps you need to follow for your bowel prep as you prepare your bowel for surgery. Clear Liquid Diet"Clear liquid" means no sediment, no fiber, and no pulp. Clear does not mean colorless.

A clear liquid diet includes:

• Water

• Popsicles-no fruit fiber

• Sparkling water or soda

• Broth or bouillon

• Crystal Light®

• Juice-no pulp (apple okay, no orange)

• Hard candy

• Coffee or tea, no creamer – sugar is OK

• Gatorade® or other clear sports drink

• No alcohol or dairy products

• Ensure Clear® (protein drink)

2 Days before Surgery

Follow the instructions on your checklist and purchase the Bisacodyl Laxative, MiraLAX®, Gatorade® or Powerade, Fleets® Phosphate Enema, Ensure Clear®

Pick up the antibiotics neomycin and metronidazole. Prescriptions for these will be sent to your preferred pharmacy.

Day before Surgery

Follow the schedule on your checklist for completing your bowel prep. Clear your schedule of other duties for the day.

Clear Liquids

Drink only clear liquids all day. Nothing solid by mouth.

Bisacodyl

Take your bisacodyl tablets in the morning.Do not chew or crush. If you take morning medicine, it is okay to take them with the bisacodyl.

Other Medicines

If you take scheduled medicines, take them at least 1 hour or more before or after drinking the prep.

Bowel Prep

Follow your ERAS Bowel Prep Checklist.You will start drinking the prep around noon (12 PM). Yo may feel some cramps in your lower abdomen while drinking the

Checklist annex

Data from the anesthesia literature have demonstrated that intake of clear fluids up until 2 hours before surgery does not increase gastric content, reduce gastric fluid pH, or increase complication rates. Thus, clear fluids should be allowed up to 2 hours before induction of anesthesia and solids up to 6 hours prior. Integration of a multidisciplinary approach is important to ensure buy-in and compliance with these guidelines from all members of the surgical team.

3.6.3 Summary and recommendation:

Mechanical bowel preparation alone with systemic antibiotic prophylaxis has no clinical advantage and can cause dehydration and discomfort and should not be used routinely in colonic surgery, but may be used for rectal surgery. There is some evidence from randomized controlled trials to support the use of a combination of MBP and oral antibiotics over MBP alone

Session 3.7 Thromboembolism prevention

Session duration: 60 Minutes

Session objective: At the end of this chapter, participants will be able to

- ✓ Describe risk factors for thromboembolism
- ✓ Explain assessments of the risks of thromboembolism
- ✓ Discuss thromboembolism prevention in ERAS perspective

3.7.1. Introduction

Activity: Individual Reflections

How often thromboembolic events occur among surgical patients?

What was your experience on perioperative thromboembolic events?

What are symptoms and signs of thromboembolism?

Time: 5 minutes

Venous thromboembolism (VTE) encompasses a spectrum of conditions ranging from asymptomatic leg vein thrombosis to fatal pulmonary embolism (PE). Surgical patients are considered to be at increased risk of VTE. It is the third most common cause of death in hospitalized patients. Non-fatal PE is associated with significant short- and long-term morbidity and represents a considerable healthcare burden.

The Virchow's triad of endothelial injury, venous stasis, and hypercoagulability, the classically described risk factors for VTE, are all present in the perioperative setting, thereby contributing to the high incidence of postoperative VTE. Thus, in patients having orthopedic surgery without thromboprophylaxis, rates of symptomatic DVT and PE are approximately 3% and 1.5% at 35 days, respectively (1).

Venous thromboembolism warrants significant attention due to its sizeable morbidity and mortality. Irrespective of the causes, the fatality rate for PE is estimated to be as high as 9%. Clot extension, embolism, post-thrombotic syndrome, and chronic thromboembolic pulmonary hypertension are additional complications of VTE that justify the implementation of tailored preventative strategies. The appropriate utilizations of perioperative VTE preventions are major pillar in the ERAS program and found to reduce the occurrence of thromboembolic events (2, 3). These preventive strategies can be broadly classified as pharmacologic and non-pharmacologic approaches. The appropriate strategies for VTE prevention must evaluate patient, surgical, and

anesthetic risks of bleeding and thrombosis. Venous thromboembolism is a common postoperative complication, but effective prophylactic strategies, including both non-pharmacologic and pharmacologic means, have reduced its incidence.

3.7.2. Risk factors for thromboembolism and assessment

Activity: Group discussion and gallery walk

What are the risk factors for thromboembolism?

How to assess risk of thromboembolism?

Time: 10 Minutes

The risk factors for thromboembolism can be classified into patient factors, surgical factors, and disease process (**Table 1**). Individualized risk assessment can be done by using different risk assessment scores. The Caprini score is a commonly used and valid scoring system to help risk assessment and guide prevention of thromboembolic events among surgical patients (**Annex 2**).

Table 1: Risk factors for venous thromboembolism

Patient factors	Surgical factors	Disease process
Age > 60 years	Any surgery > 90 minutes	Active cancer
Obesity (BMI > 30 Kg/m²)	Pelvic and lower limb surgery > 60 minutes	Cancer treatment
Existing varicose veins with phlebitis	Acute intra-abdominal conditions	Thrombophilia
Dehydration	Significant immobility expected	Heart disease
Family history of VTE	Critical care unit admission	Metabolic/endocrine conditions
Pregnancy and puerperium		Acute infectious disease
Hormone replacement therapy		Inflammatory conditions
Estrogen-based contraceptives		

BMI: Body Mass Index, VTE: Venous Thromboembolism

3.7.3. Prevention of thromboembolism

One of the basic bundles of ERAS is the prevention of thromboembolic events. Thromboembolism prevention can be achieved by various methods that are broadly classified as non-pharmacologic and pharmacologic. The prevention of thromboembolic events should be guided by the risk of occurrence of the events and bleeding (**Annex 3**).

Case 1

Case 2

Case 3

Recognition of thromboembolic events

References

1. Bell BR, Bastien PE, Douketis JD. Prevention of venous thromboembolism in the Enhanced Recovery After Surgery (ERAS) setting: an evidence-based review. Canadian Journal of Anesthesia. 2015;62(2):194.

2. Taylor JS, Iniesta-Donate M, Cain K, Lasala J, Mena G, Zorilla Vaca AS, et al. Rate of venous thromboembolism in gynecologic patients on an ERAS pathway. American Society of Clinical Oncology; 2021.

3. Vendler M, Haidari T, Waage J, Kleif J, Kristensen B, Gögenur I, et al. Incidence of venous thromboembolic events in enhanced recovery after surgery for colon cancer: a retrospective, population-based cohort study. Colorectal Disease. 2017;19(11):O393-O401.

Session 3.8 General and regional anesthesia management in ERAS

Duration: 60 Minutes

Session Objectives

- Identify General anesthesia management principles in ERAS
- Describe regional Anesthesia management in ERAS

Activity: Group Discussion

How do you select better technique of Anesthesia for ERAS?

Time: 10 Minutes

3.8.1 Introduction

There is little evidence to favour one anaesthetic technique over another but the general principles of enhanced recovery support the use of medications that have minimal post-operative hang-over and effects on gastrointestinal motility. Anesthetic management in the perioperative period plays a critical role in the ERAS system implementation. By facilitating the implementation of these protocols through communication and collaboration between personnel in the different perioperative periods, the anesthesia team significantly improves patient satisfaction with their smooth transition from the preoperative to intraoperative to the postoperative period. In addition, regional anesthesia practice including neuraxial block, chest wall blocks and abdominal field blocks are majorly advocated in ERAS (3).

3.8.2 General anesthesia management principles in ERAS

General Anesthesia management focuses on, fast recovery of patients postoperatively by choosing shorter-acting premedications and using reduced levels of volatile anesthetics, or using total intravenous anesthesia (TIVA), minimizing the use of opioids and adequate pain management, maintaining euvolemia, normothermia, and administering prophylactic antiemetics. While enhanced recovery protocols are being implemented at an increasing rate and growing in popularity, we cannot overlook persistent patient safety threats that could also prove to be significant impediments to such programs. As ERAS protocols rely upon well-established evidence, well-described strategies to avoid postoperative residual weakness should be incorporated as the perioperative community continues efforts to advance patient safety and improve outcomes during neuro muscular blocking agents' usage (4).

3.8.3 Regional Anesthesia management in ERAS

Regional anaesthesia (RA) techniques, including neuraxial and peripheral nerve block, can provide many benefits for patients in the perioperative period. These benefits include a decrease in postoperative pain (subsequently reducing opioid consumption and associated adverse effects), decrease in nausea and vomiting, improvement in mobilization and recovery of gastrointestinal function, decrease in length of stay (LOS), reduction in surgical stress response, and potentially, reduction in morbidity and mortality. They are therefore commonly used to improve quality of patient care and have also been used as a key component of many enhanced recovery protocols (5).

Activity: Individual reflection Describe monitoring methods of neuromuscular blocks and regional anesthesia/analgesia Time: 25 minutes

Session Summary

ERAS (Enhanced Recovery After Surgery) is an approach to anesthesia management which focuses on preoperative patient optimization, early epidural analgesia, avoidance of multiple potent sedatives, and improved postoperative patient management. It involves the use of multimodal analgesia, accelerated recovery protocols, and involvement of the patient in the care process. The goal of this approach is to reduce postoperative pain and nausea, shorten length of stay, and decrease complications.

Session 3.9: ERAS in the Post-Anesthetic Care Unit (PACU)

Session duration: 60 Minutes

Session objective: At the end of this session, participants will be able to;

- ✓ Describe care protocols in the PACU in ERAS perspectives

3.9.1. Introduction

Activity: Small group discussion and reflections

How patient care should be in the PACU according to ERAS protocols?

Time: 10 minutes

In recent times, the PACU has seen the incorporation of advanced technology, such as patient monitoring systems, improved pain management techniques, and increased staffing levels. The PACU has contributed to the current standard of care, which is focused on providing safe and effective recovery for patients after they have received anesthesia. The ERAS protocols involve the implementation of various evidence-based practices and protocols to improve the surgical experience for patients and optimize their recovery.

During postoperative care in the PACU, patients are closely monitored and given appropriate support and treatment to promote their recovery. This includes early mobilization, adequate pain control, effective nausea and vomiting management, and the provision of appropriate nutrition. All of these lead to achieving the goal of ERAS that are reducing the length of hospital stay, minimize complications and improve the patient's overall well-being.

3.9.2 Main activities in PACU

In the post-anesthetic care unit (PACU), protocols are used to ensure the safe and effective management of patients after surgery. Here are some of the key components that are typically included in PACU protocols for patients following an ERAS procedure:

1. Monitoring: Vital signs such as blood pressure, heart rate, respiratory rate, and oxygen saturation are closely monitored, and appropriate interventions are taken if necessary.
2. Pain management: Patients are assessed for pain and given appropriate pain management, such as non-opioid analgesics, regional anesthesia, or nerve blocks, as per the ERAS protocol.

3. Nausea management: Patients are assessed for nausea and given appropriate treatment, such as antiemetics, as per the ERAS protocol.
4. Mobilization: Patients are encouraged to get out of bed and move around as soon as possible, as part of their postoperative care.
5. Nutrition: Patients are encouraged to start oral intake as soon as possible and are provided with appropriate nutrition, such as clear fluids or a light meal, as per the ERAS protocol. Respiratory management: Patients may receive respiratory support, such as supplemental oxygen, airway suctioning, or mechanical ventilation, as per the ERAS protocol.
6. Wound management: Patients may receive wound care, such as dressing changes or wound irrigation, as per the ERAS protocol.
7. Medication administration: Patients may receive medications, such as antibiotics, anticoagulants, or pain management medications, as per the ERAS protocol.
8. Monitoring for complications: Patients are monitored for potential complications, such as bleeding, wound infection, or adverse reactions to medications, and appropriate interventions are taken if necessary.
9. Psychological support: Patients may receive psychological support, such as counseling or relaxation techniques, to manage anxiety or stress related to the surgery, as per the ERAS protocol.
10. Sedation management: Patients may receive medications to manage agitation, anxiety, or pain, as per the ERAS protocol.
11. Oxygen therapy: Patients may receive supplemental oxygen to maintain adequate oxygen saturation, as per the ERAS protocol.
12. Monitoring for readmission criteria: Patients are monitored for criteria that may require readmission to the hospital, such as worsening vital signs or new symptoms, as per the ERAS protocol.
13. Monitoring for discharge criteria: Patients are monitored for discharge criteria, such as stable vital signs, appropriate pain management, adequate oral intake, and readiness for discharge, as per the ERAS protocol.
14. Discharge planning: Patients are assessed for readiness to leave the PACU, and appropriate discharge instructions and follow-up appointments are arranged.

15. Discharge coordination: The PACU staff coordinates the patient's discharge, including arranging transportation, arranging for equipment or supplies needed at home, and communicating with other healthcare providers involved in the patient's care.
16. Discharge instructions: Patients receive comprehensive discharge instructions, including information on pain management, wound care, activities of daily living, and follow-up appointments.
17. Family and caregiver education: Family members and caregivers may be educated on the patient's condition, recovery process, and postoperative care instructions, as per the ERAS protocol.
18. Handoff communication: The PACU staff communicates with the patient's next level of care, such as the ward or the intensive care unit, to ensure seamless and coordinated care.
19. Quality improvement and patient safety: The PACU protocols may include elements to continuously monitor and improve the quality of care and patient safety, such as regular audits and incident reporting.

These PACU protocols are designed to ensure that patients receive safe and effective care after surgery, and to promote their recovery. By following these protocols, healthcare providers can ensure that patients have a smooth and successful recovery from their surgery.

Activity: Small group discussion and presentation

Discuss discharge criteria from the PACU.

Discuss pain management in the PACU.

Discuss nausea and vomiting management in the PACU.

Time: 20 minutes

Summary

- The PACU protocols are comprehensive guidelines that ensure the safe and effective management of patients after surgery.
- These protocols outline key components such as patient assessment, pain management, vital sign monitoring, medication administration, and discharge criteria. Additionally, PACU protocols may include elements such as wound management, respiratory support, psychological support, readmission criteria, mobility and rehabilitation, and follow-up care.
- The goal of PACU protocols is to ensure that patients receive high-quality, comprehensive, and coordinated care during their postoperative recovery period, which promotes their overall well-being and successful recovery from surgery.

ERAS in the PACU/recovery room

Session 3.10 Epidural and caudal blocks

Duration 50 minutes

Session objectives
Perform epidural and caudal blocks in accordance with ERAS recommendations

3.10.1 Introduction

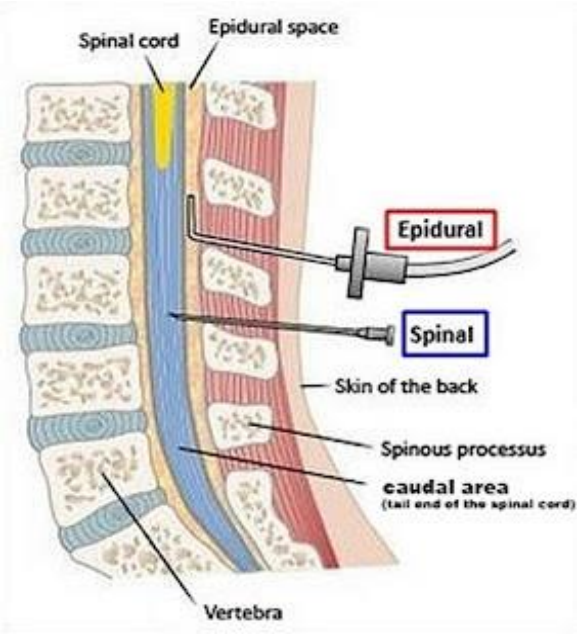
Activity : individual reflections
Describe the importance of epidural and caudal blocks as ERAS protocol.
Time 10 minutes

Epidural analgesia has been advocated for postoperative pain control, for the management of the neuroendocrine response, and for its effects on postoperative stay. A retrospective study and two randomized studies both reported that epidural analgesia outside an ERAS program improved pain control in laparoscopic colorectal surgery. Continuous Thoracic Epidural Aanalgesia with local anesthetics (bupivacaine, ropivacaine) is likely the core of neuraxial analgesia in ERAS protocols providing effective postoperative analgesia in open abdominal surgery with uncertain evidence in laparoscopic procedures.

3.10.2 Techniques of epidural block

Activity: group discussion and reflection
Describe the structures the needle goes through until it reach epidural space.
10 minutes

In the medial approach, the site of the insertion of the needle is between the spaces created by the vertebral spinous processes. Upon locating the desired spot, lidocaine 1% must be injected into the skin and underlying tissues to decrease the discomfort with the advancement of the epidural needle. Once achieving local anesthesia, the epidural needle must be advanced with its stylet in place and with its bevel point cephalad; this will ultimately contribute to the proper location of the epidural catheter. The epidural needle must be advanced through the skin, subcutaneous tissue, supraspinous, and interspinous ligaments. Once there, the stylet must be removed, and the Loss of Resistance syringe (filled up with saline, air, or both) must be attached to the needle. The needle must be advanced while applying pressure to the plunger. Once the ligamentum flavum is pierced, a loss in resistance will be noted; this is the epidural space, and 5 to 10 cc of saline may be injected to expand the epidural space; this may decrease the risk of vascular injury.



3.10.3 **Techniques of caudal block**

Caudal anesthesia is a variation of epidural anesthesia widely used in childhood. It is useful for interventions under the umbilicus line (circumcision, herniotomy, orchiopexy). The indications and contraindications match the ones described above.

The patient positioning is in lateral decubitus, fetal position, the area sterile prepped, and the hiatus sacralis located at the vertex of the sacrum, between the cornua sacra. The sacrococcygeal membrane must be pierced with a 22 to 25 G venous catheter at a 45-degree angle relative to the body's longitudinal axis. The catheter must be advanced until the loss of resistance is felt. No additional advance must be attempted since the distance between the sacrococcygeal membrane and the caudal end of the dural sac may be less than 10 mm. If CSF or blood is noted, the catheter must be repositioned. Once properly located, a test dose with Epinephrine 0.5 mcg/kg may be performed to rule out an intravascular injection.

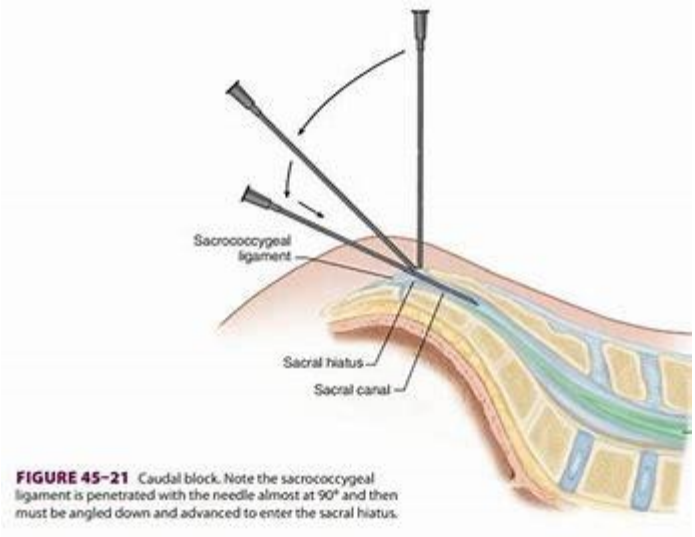


FIGURE 45–21 Caudal block. Note the sacrococcygeal ligament is penetrated with the needle almost at 90° and then must be angled down and advanced to enter the sacral hiatus.

3.10.4 Assessing block and troubleshooting

Activity: group discussion and reflection Describe how to asses the level of the block and guarantee surgical anesthesia. 10 minutes

Immobility is essential to enable surgery. General anesthesia guarantees this by making patients unconscious and unresponsive to a surgical stimulus. If an epidural achieves dense motor block this may keep the lower half immobile. However, regional techniques ultimately rely on a co-operative patient to stay still. Without dense motor block legs may move involuntarily and upper half movement may interfere with surgery indirectly.

Pre-procedure testing should suggest that pain is absent in the surgical field

Session 3.11: Chest wall blocks in ERAS

Session duration: 60 Minutes

Session objective: At the end of this session, participants will be able to;

- ✓ Describe common chest wall blocks in ERAS
- ✓ Recognize indications of chest wall blocks
- ✓ Discuss the complications of chest wall blocks

3.11.1. Introduction

Activity: Small group discussion and presentation

What are the commonly used chest wall blocks and their benefits in ERAS?

What are the indications of chest blocks?

Time: 15 minutes

Systemic analgesia is complemented by the use of local anesthetic injections or infusions around specific peripheral nerves or nerve plexus, into tissue at the site of incision, into joint spaces, peritoneal spaces, tissue planes, and via epidural or intrathecal routes.

Chest wall blocks are key components of ERAS protocols that are used to provide pain relief after surgery. They are typically used for thoracic and abdominal surgeries. Chest wall blocks involve injecting a local anesthetic into the intercostal nerves, which are located between the ribs. Chest wall blocks can provide effective pain relief for up to 24 hours, allowing patients to be more comfortable and mobile in the early postoperative period. Chest wall blocks can also reduce the need for opioid medications, which can help reduce the risk of opioid-related side effects such as nausea, vomiting, constipation, and respiratory depression. Chest wall blocks are safe and effective way to provide pain relief for patients undergoing thoracic and abdominal surgery.

3.11.2 Paravertebral block

Paravertebral block is a type of regional anesthesia used in ERAS protocols that reduce postoperative pain and opioid consumption, as well as to reduce the length of hospital stay. The block is administered by injecting a local anesthetic into the paravertebral space, which is located between the vertebrae. This technique is used

to provide postoperative pain relief for patients undergoing thoracic or abdominal surgery. The paravertebral block is performed using ultrasound guidance or nerve stimulation.

In ERAS protocols, paravertebral blocks can be used in combination with other forms of anesthesia, such as general anesthesia, to provide comprehensive pain control during and after surgery. They can also be used alone for specific types of thoracic or abdominal procedures.

Benefits of paravertebral blocks in the context of ERAS include improved pain control, reduced need for opioid medications, reduced stress response, shorten hospital stay, and improved patient satisfaction. They also have a relatively low risk of side effects compared to other forms of anesthesia such as epidural anesthesia.

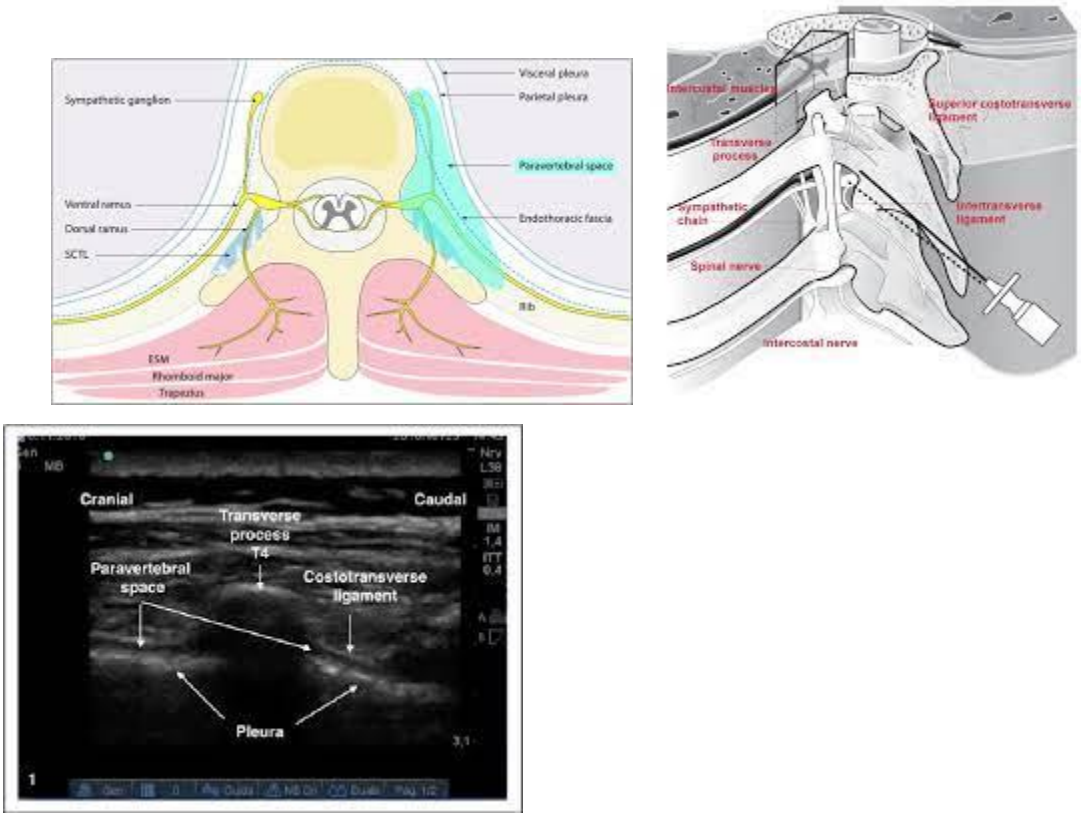


Figure 1: Paravertebral block

3.11.3 Intercostal/subcoastal blocks (ICIS)

Intercostal/subcostal blocks (ICIS) are types of regional anesthesia that is used to provide pain relief during and after surgery. The ICIS blocks are administered by injecting a local anesthetic into the intercostal/subcostal nerves, which are located between the ribs. These blocks are usually used in combination with general

anesthesia, and it can be used for a variety of surgery, including abdominal, thoracic, and breast surgeries. It can also be used for post-operative pain relief. The blocks can be administered either as a single injection or as a continuous infusion.

When the ICSC blocks is used in ERAS, it is important to ensure that the patient is adequately monitored and that the anesthetic is administered correctly. The block should be administered in a way that minimizes the risk of complications, such as nerve damage or infection.

In addition to providing pain relief, the intercostal block can also help to reduce the amount of general anesthesia that is needed, which can help to reduce the risk of side effects and complications associated with general anesthesia. It can also help to reduce the amount of time that the patient needs to stay in the hospital after surgery. Overall, intercostal/subcoastal blocks in ERAS can be an effective way to provide pain relief and reduce the amount of general anesthesia that is needed. However, it is important to ensure that the block is administered correctly and that the patient is adequately monitored.

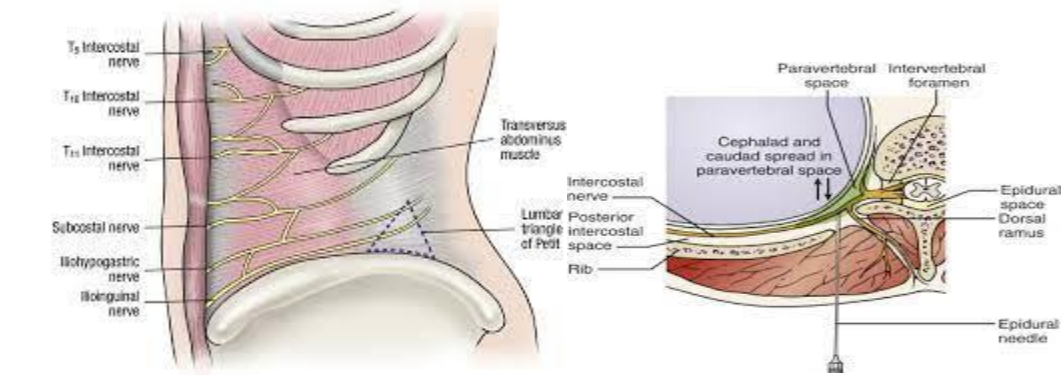


Figure 2: Intercoastal/subcoastal blocks (ICIS)

3.11.4. Erector Spinae block (ESB)

Erector spinae blocks (ESB) are a type used in ERAS protocols to provide postoperative analgesia and can be used in combination with other forms of anesthesia and analgesia. The ESB are typically placed in the

thoracic or lumbar region of the spine, and can be used to block nerve pathways that are responsible for transmitting pain signals from the surgical site.

Erector spinae blocks (ESB) can provide effective pain relief for up to 24 hours, and can be used to reduce the need for opioid analgesics, which can reduce the risk of opioid-related side effects. These blocks can also be used to reduce the risk of postoperative nausea and vomiting, as well as reduce the length of hospital stay. ESB can be used in combination with other forms of regional anesthesia, such as epidural or paravertebral blocks, to provide more comprehensive pain relief.

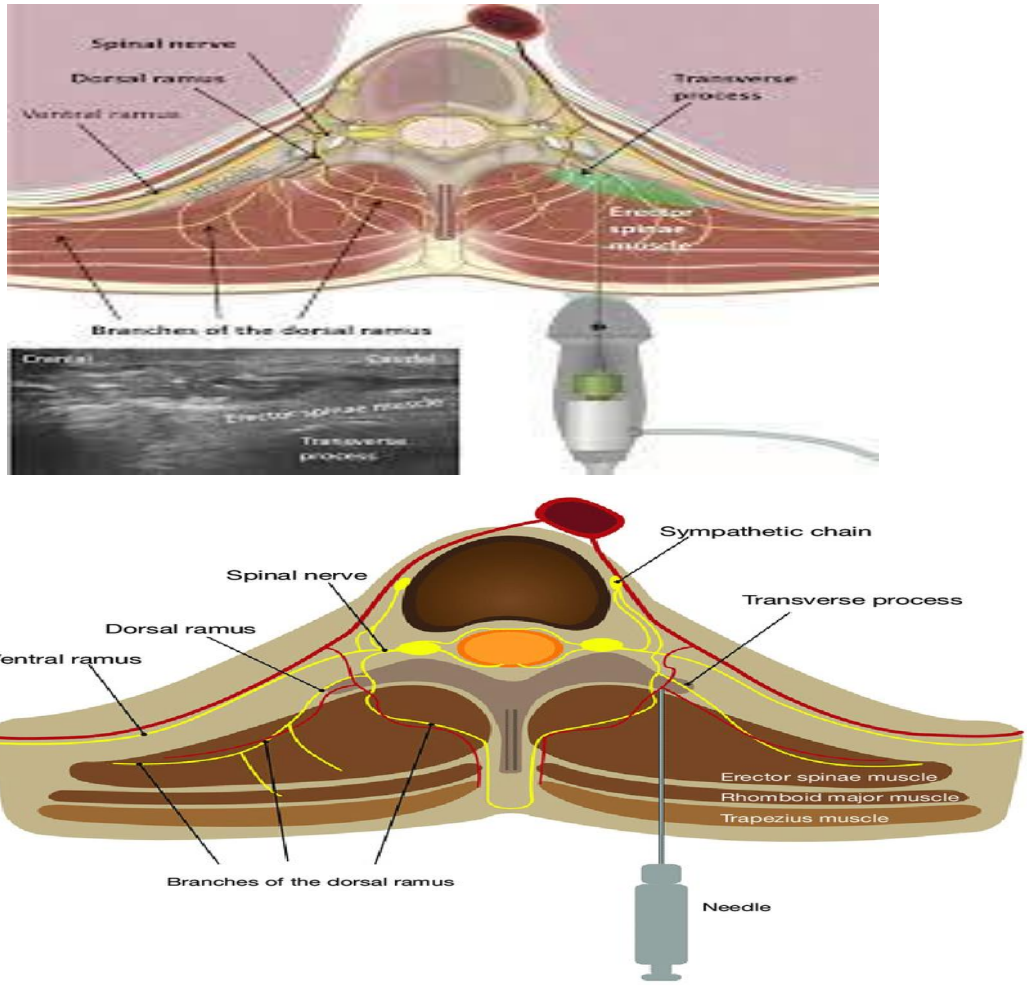


Figure 3: Erector spinae block (ESB)

Activity: Small group discussion and gallery walk

What are the complications of chest wall blocks?

Time: 20 minutes

Summary

- Chest wall blocks, as the important components of ERAS protocols, reduce complications, and improve the quality of recovery after surgery.
- Chest wall blocks improve patient outcome when included in multimodal pain management.

Session 3.12: Abdominal field blocks in ERAS

Session duration: 60 Minutes

Session objective: At the end of this session, participants will be able to;

- ✓ Describe common abdominal field blocks in ERAS
- ✓ Recognize indications of abdominal field blocks
- ✓ Discuss the complications of abdominal field blocks

3.12.1. Introduction

Activity: Small group discussion and Reflection

What are the benefits of abdominal field blocks in ERAS?

What are the indications of abdominal field blocks

What are the complications of abdominal field blocks

Time: 15 minutes

Systemic analgesia is complemented by the use of local anesthetic injections or infusions around specific peripheral nerves or nerve plexus, into tissue at the site of incision, into joint spaces, peritoneal spaces, tissue planes, and via epidural or intrathecal routes.

Abdominal field blocks are a type of regional anesthesia used in Enhanced Recovery After Surgery (ERAS) protocols. They are used to provide pain relief for abdominal surgeries, such as laparoscopic and open abdominal procedures. This blocks the transmission of pain signals from the surgical site to the brain, providing a more comfortable recovery experience for the patient.

The use of abdominal field blocks in ERAS protocols has been shown to reduce post-operative pain and opioid use, improve patient satisfaction, and reduce hospital length of stay. Additionally, the use of regional anesthesia in ERAS protocols has been associated with improved outcomes in terms of reduced surgical site infections, improved wound healing, and reduced post-operative nausea and vomiting.

Overall, the use of abdominal field blocks in ERAS protocols is an important part of providing safe and effective pain relief for patients undergoing abdominal surgery. The benefits of using regional anesthesia in ERAS protocols are numerous, and should be considered when planning for a patient's surgical procedure.

Activity: Small group discussion and presentation

Discuss the techniques of common abdominal field blocks.

Time: 20 minutes

3.12.2. Trans-abdominis plane block (TAP)

Trans-abdominis plane (TAP) block is a type of regional anesthesia used in Enhanced Recovery After Surgery (ERAS) protocols. It is a form of nerve block that is used to provide pain relief to the abdomen and lower back. It is typically used in combination with other forms of anesthesia, such as general anesthesia or spinal anesthesia, to provide a more comprehensive pain relief regimen.

The TAP block is performed by injecting a local anesthetic, such as bupivacaine, into the transverse abdominis plane (TAP) of the abdomen. This plane is located between the internal oblique and transverse abdominis muscles. The injection is typically done in a fan-like pattern, with multiple injections placed along the length of the TAP plane. The local anesthetic works by blocking the nerves that supply sensation to the abdomen and lower back, providing pain relief to the area. The TAP block is a relatively safe procedure, with few reported complications. It has been shown to provide effective pain relief for up to 72 hours after surgery, making it an ideal choice for ERAS protocols. Additionally, it has been shown to reduce opioid consumption and improve patient satisfaction after surgery.

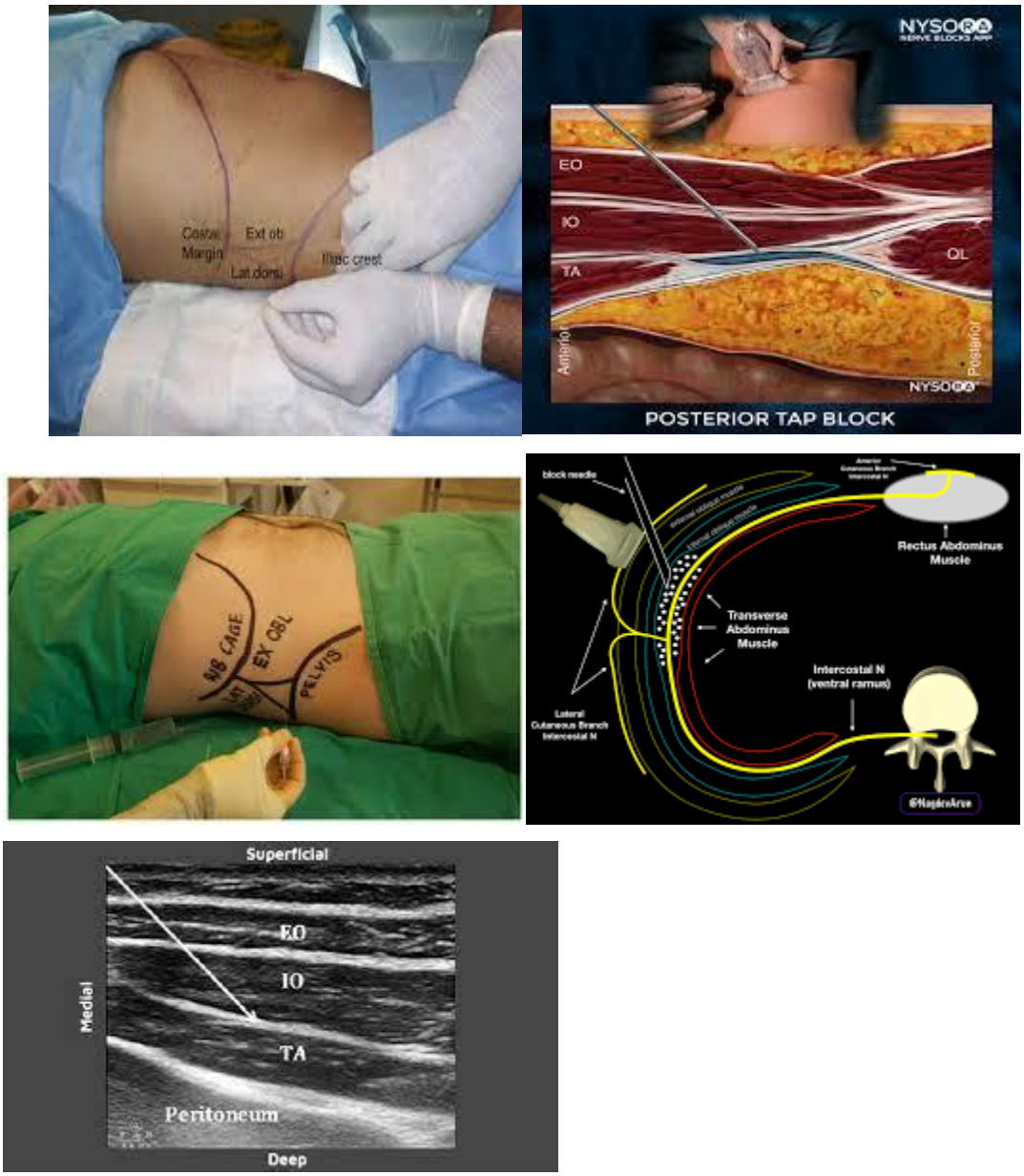


Figure 1: Trans-abdominis plane block

3.12.3. Ilio-inguinal-ilio-hypogastric block (ILIG)

The Ilio-inguinal-ilio-hypogastric block is a type of nerve block that is used to provide pain relief in the lower abdomen, groin, and upper thigh. It is a type of regional anesthesia that is used to provide pain relief during surgical procedures, such as hernia repair, appendectomy, and inguinal hernia repair. In the era ERAS, ILIG block is becoming increasingly popular as a means of providing pain relief. This is due to their ability to provide effective, long-lasting pain relief with minimal side effects.

Ilio-inguinal-ilio-hypogastric block (ILIG) can be used as a single-shot or continuous infusion, depending on the patient's needs. The block is performed by injecting a local anesthetic solution into the area around the ilioinguinal and iliohypogastric nerves. These nerves are located in the lower abdomen, near the inguinal ligament.

In the era of ERAS, ILIG block is becoming increasingly popular due to their ability to provide effective, long-lasting pain relief with minimal side effects. They can also be used in combination with other forms of anesthesia, such as general anesthesia, to provide a more comprehensive approach to pain relief. Additionally, ilioinguinal blocks can be used to reduce the need for opioids, which can help reduce the risk of opioid-related side effects and addiction.

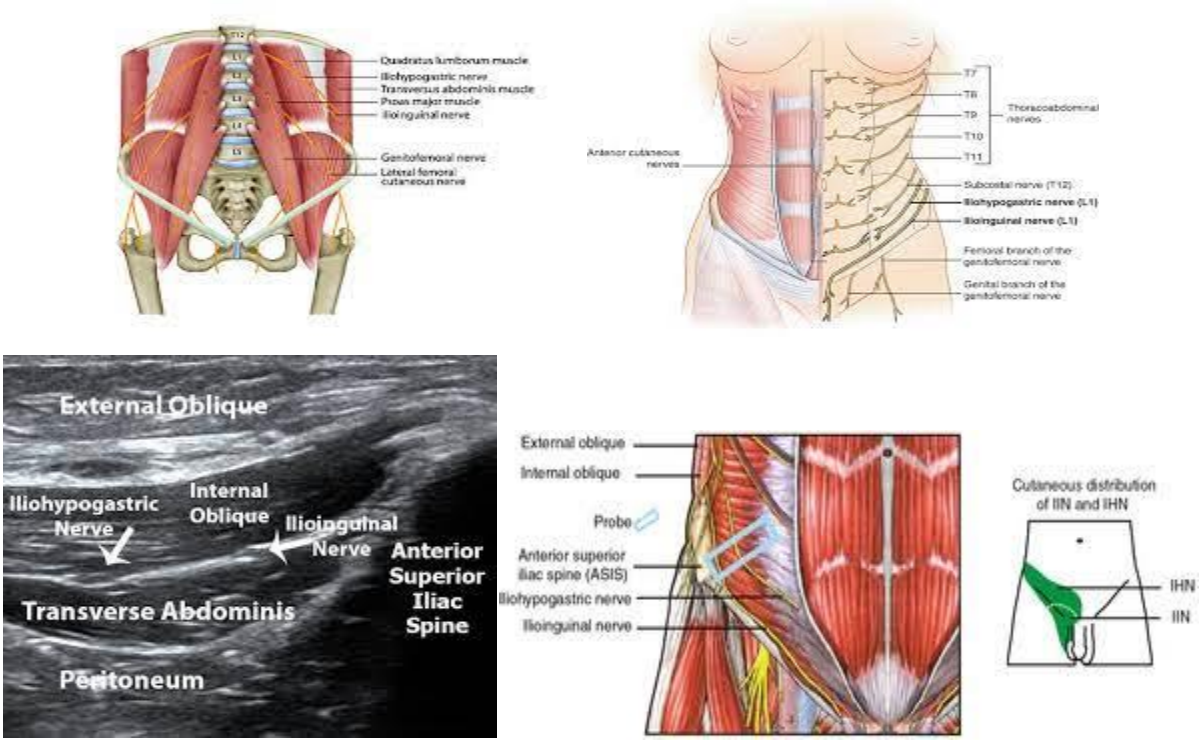


Figure 2: Ilio-inguinal-ilio-hypogastric block

3.12.4. Quadratus Lumborum Block (QLB)

A quadratus lumborum block (QLB) is a type of regional anesthesia that is used in the perioperative setting to provide pain relief for patients undergoing abdominal, pelvic, and lower extremity surgeries. The QLB is a nerve block that targets the quadratus lumborum muscle, which is located in the lower back and is responsible for stabilizing the spine and hip joint. This block is used to provide pain relief for patients undergoing abdominal, pelvic, and lower extremity surgeries. It is also used to reduce the need for opioid analgesics, which can have

significant side effects. The QLB is performed by injecting a local anesthetic, such as lidocaine or bupivacaine, into the quadratus lumborum muscle. The injection is typically done under ultrasound guidance, which allows the anesthesiologist to accurately target the muscle. The block can be performed as a single injection or as a continuous infusion.

The effects of the block typically last for several hours, but can be extended with a continuous infusion. The QLB is a safe and effective technique that has been used in the perioperative setting for several years. It has been shown to be effective in reducing postoperative pain and opioid use in patients undergoing abdominal, pelvic, and lower extremity surgeries. It is also a cost-effective technique that can reduce the need for opioid analgesics, which can be expensive.

The QLB is an important part of ERAS protocols, which are designed to reduce the length of hospital stays and improve patient outcomes. The QLB can be used to reduce pain and opioid use, which can help to reduce the length of hospital stays and improve patient outcomes.

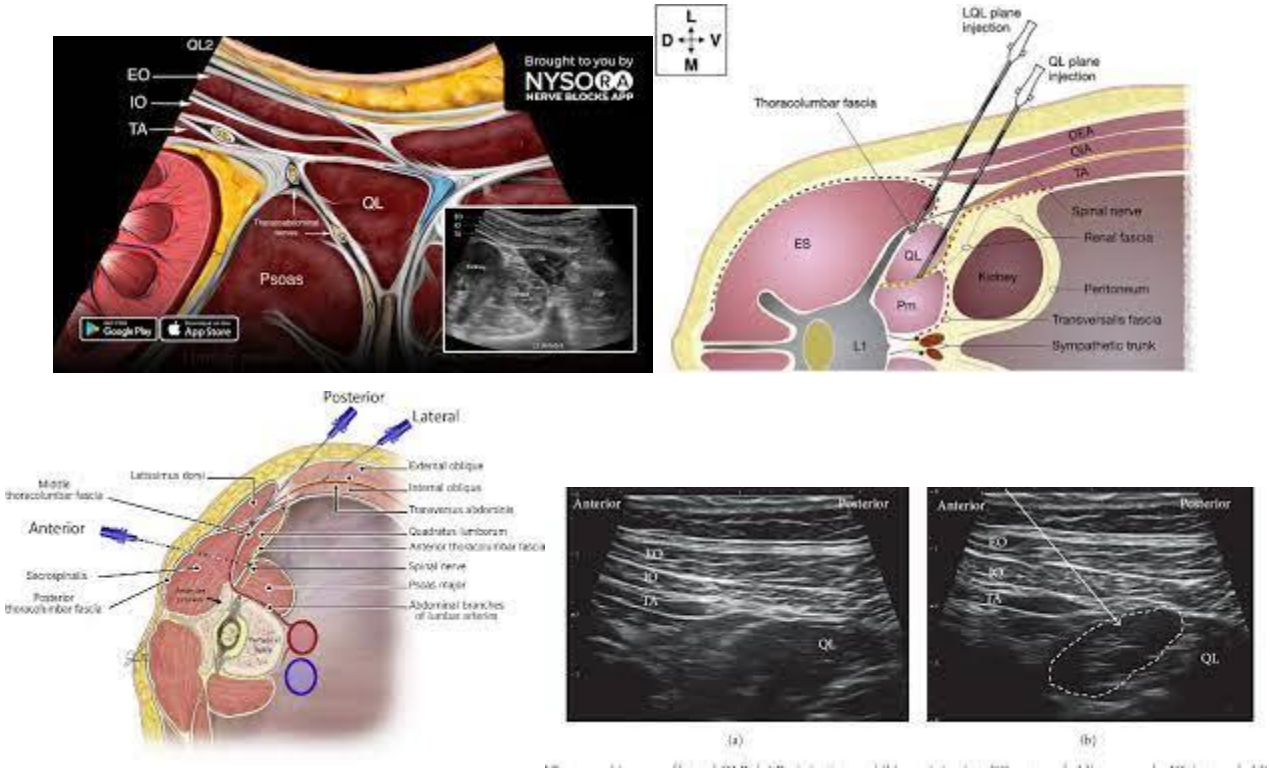


Figure 3: Quadratus lumborum block

3.12.5. Rectus Sheath Block

The use of rectus sheath block in ERAS protocols is an increasingly popular practice. The rectus sheath block is an injection of local anesthetic into the rectus sheath, which is the area around the rectus abdominis muscle. It has been found to be beneficial for patients undergoing abdominal surgeries, as it helps to reduce post-operative pain, decrease opioid requirements, and decrease length of hospital stay. The rectus sheath block has been found associated with decreased opioid doses taken in the first 24 hours after surgery, decreased pain levels, and reduced lengths of stay in the hospital. It is a safe, effective, and cost-efficient option for providing anesthesia during abdominal surgeries, and should be considered as part of an ERAS approach.

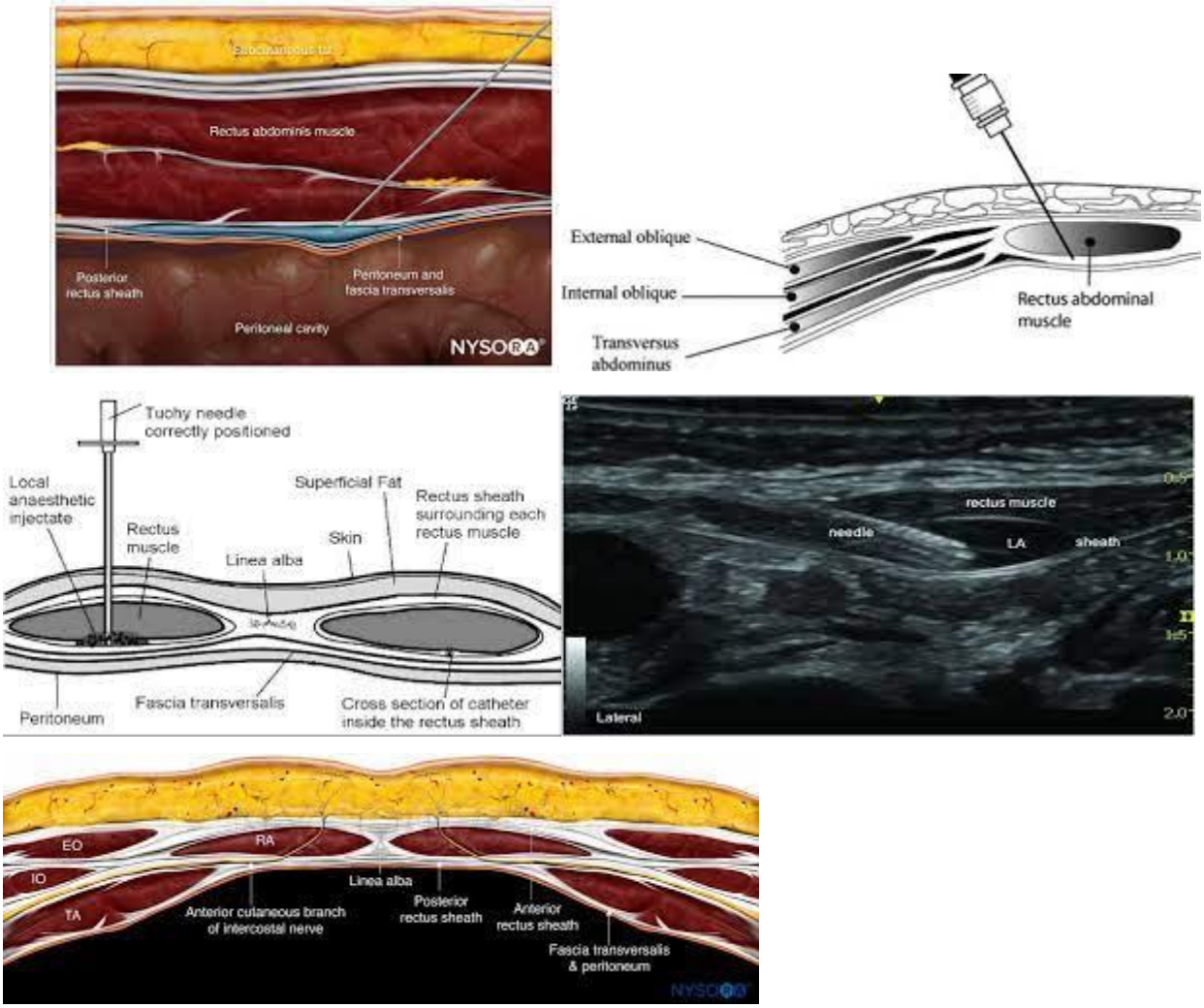


Figure 2: Rectus sheath block

3.12.6 Assessment and monitoring of abdominal field blocks

To assess efficacy, various methods such as patient-reported pain scores, vital signs, and use of supplemental pain medication can be used. Additionally, nerve stimulation techniques can be employed to confirm proper placement of the block and ensure adequate spread of the anesthetic.

For safety, monitoring of vital signs and potential adverse effects such as nausea, vomiting, and respiratory depression is important. The use of ultrasound or nerve stimulation can also minimize the risk of nerve injury. Additionally, proper sterile technique and avoiding injection into blood vessels can reduce the risk of infection.

In conclusion, assessment of the efficacy and safety of abdominal field blocks is crucial to ensure optimal patient outcomes during surgical procedures. Regular monitoring and proper technique can help minimize potential adverse effects and ensure effective pain management.

Activity: group discussion and reflection

Describe assessment and monitoring modalities of abdominal field blocks

What are the importance of assessing the abdominal nerve blocks

Time 5 minutes

3.12.7 Session Summary

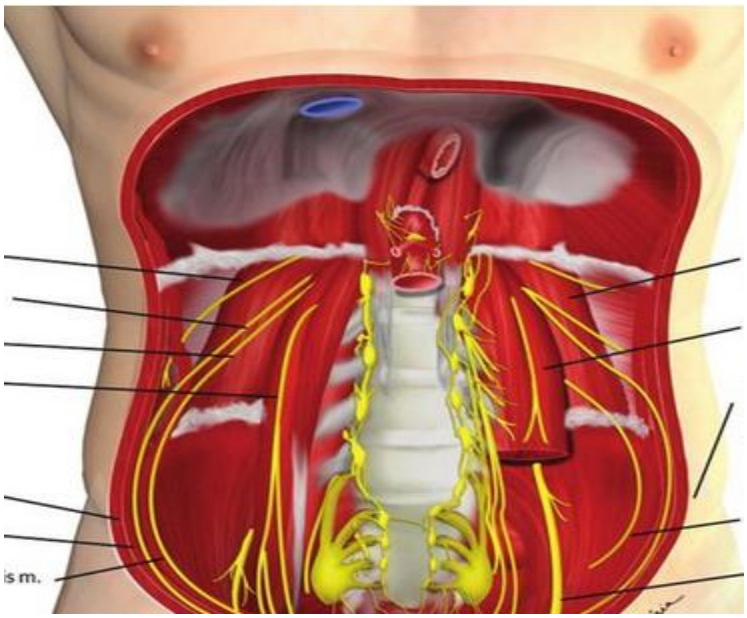
Abdominal field blocks are a part of an Enhanced Recovery After Surgery (ERAS) program. This program is designed to help patients recover faster and more comfortably after abdominal surgery. Abdominal field blocks involve the injection of local anesthetic around the surgical site to provide pain relief. This type of pain relief is beneficial as it reduces the need for opioid medications, which can have side effects.

- The implementation of ERAS protocols in different surgical procedures reduces overall complications and recovery time.
- Multimodal analgesia strategies with regional anesthesia techniques are some of the key interventions contributing to the improvement in postoperative outcomes.

Activity: Group discussion and Reflection

1. Name the nerves illustrated in the picture below
2. Describe types of nerve blocks to address these nerves

   Time: 20 minutes



Annex 1: Instructional Content Analysis

Title of the training: **Enhanced Recovery After Surgery (ERAS)**

Job/service to improved /enhanced: Perioperative Care Service

Target audience: Surgeons, Anesthetists, Nurses, Clinical Pharmacists, Nutritionists, Physiotherapists, Operation Room Heads, Hospital and Healthcare Administrators

Major Task	Knowledge Analysis	Skill and Attitude Analysis
Patient Education and Multidisciplinary Optimization	➢ Educate patients on to dos, not to dos, and adherence ➢ Effective communication ➢ Patient rights ➢ Perioperative risk analysis ➢ Informed consent ➢ Multidisciplinary optimization of common comorbidities ➢ Impacts and management of preoperative anxiety	➢ Deliver appropriate preoperative patient education ➢ Effective communication skills ➢ Receive patient feedback ➢ Obtain informed consent ➢ Provide multidisciplinary preoperative optimization ➢ Manage preoperative anxiety
Nutritional Assessment and Optimization	➢ The impact of nutritional status on surgical outcome ➢ Preoperative nutritional assessment ➢ Nutritional supplement and optimization	➢ Assess nutritional status ➢ Provide nutritional supplement and optimization
Pain Management	➢ Basic physiology of acute pain ➢ Impacts of uncontrolled acute pain	➢ Assess acute pain ➢ Provide optimal acute pain management

	➢ Acute pain assessment ➢ Multimodal analgesia ➢ Preemptive analgesia	➢ Provide multimodal analgesia ➢ Administer preemptive analgesia
Fluid Management	➢ Perioperative hydration assessment and monitoring ➢ Goal directed perioperative fluid therapy	➢ Assess and monitor hydration status ➢ Provide goal directed fluid therapy
Postoperative Nausea and Vomiting (PONV) Prevention	➢ Basic pathophysiology of PONV ➢ Impacts of PONV ➢ Assess the risk of having PONV ➢ Prevention and management of PONV	➢ Assess the risk of having PONV ➢ Prevent PONV ➢ Manage PONV
Manage Bowel Preparation	➢ Impacts of bowel preparation ➢ Techniques of bowel preparation ➢ ERAS recommendations on bowel preparation	➢ Provide appropriate bowel preparation ➢ Implement ERAS recommendations during bowel preparation
Thromboembolism Prevention	➢ Assess the risk of the occurrence of thromboembolic events ➢ Prevention of thromboembolic events ➢ Management of thromboembolic events	➢ Assess the risks for thromboembolic events ➢ Prevent thromboembolic events ➢ Manage thromboembolic events
Regional Anesthesia in ERAS	➢ Spinal and epidural anesthesia in ERAS ➢ Chest wall blocks (including erectus spinae block)	➢ Perform neuraxial and field blocks in accordance with ERAS recommendations

	➢ Abdominal wall blocks (including quadratus lumborum block and transversalis fascia block)	
Manage and Reverse Neuromuscular Blockade	➢ Impacts of neuromuscular blockade on recovery ➢ Reversal of neuromuscular blockade	➢ Appropriate administration and monitoring of neuromuscular blockade ➢ Appropriate reversal of neuromuscular blockade
Mange Indwelling Catheters and Drains	➢ Impacts of indwelling catheters and drains ➢ ERAS recommendations on the use of catheters and drains	➢ Appropriate use of catheters and drains according to ERAS recommendations
Use of Opioids	➢ Opioid pandemics ➢ Side-effects of opioids ➢ Opioid sparing strategies ➢ Appropriate utilization of opioids	➢ Reduce side-effects of opioids ➢ Apply opioid-sparing strategies ➢ Safe utilization of opioids
Audit and Predictive Analysis	➢ Audit in ERAS ➢ Predictive analysis	➢ Conduct audit in ERAS ➢ Perform predictive analysis
Multidisciplinary Communication and Education	➢ Effective professional communication ➢ Multidisciplinary knowledge transfer	➢ Perform effective professional communication ➢ Facilitate multidisciplinary knowledge transfer
Implementation of ERAS Programs	➢ Leadership in ERAS programs	➢ Implement ERAS programs